Chapter 4

Questions on Lesson Three

- The Oscillating Circuit.
- The Resonant Circuit.

Interactive test



The questions signed by * are answered in detail. Understand Apply Higher Order Thinking Skills

Important guidelines

- **In case of a resonance circuit:**

$X_L = X_C$, $V_L = V_C$, $V = V_R$

Minimum impedance: $Z = R$

Maximum current intensity: $I = \frac{V}{R}$

The total voltage and the current are having the same phase : $\theta = 0°$

The resonance frequency of the circuit: $f = \frac{1}{2\pi\sqrt{LC}}$

To compare between the frequency of resonance of two circuits: $\frac{f_1}{f_2} = \sqrt{\frac{L_2C_2}{L_1C_1}}$

First Multiple choice questions

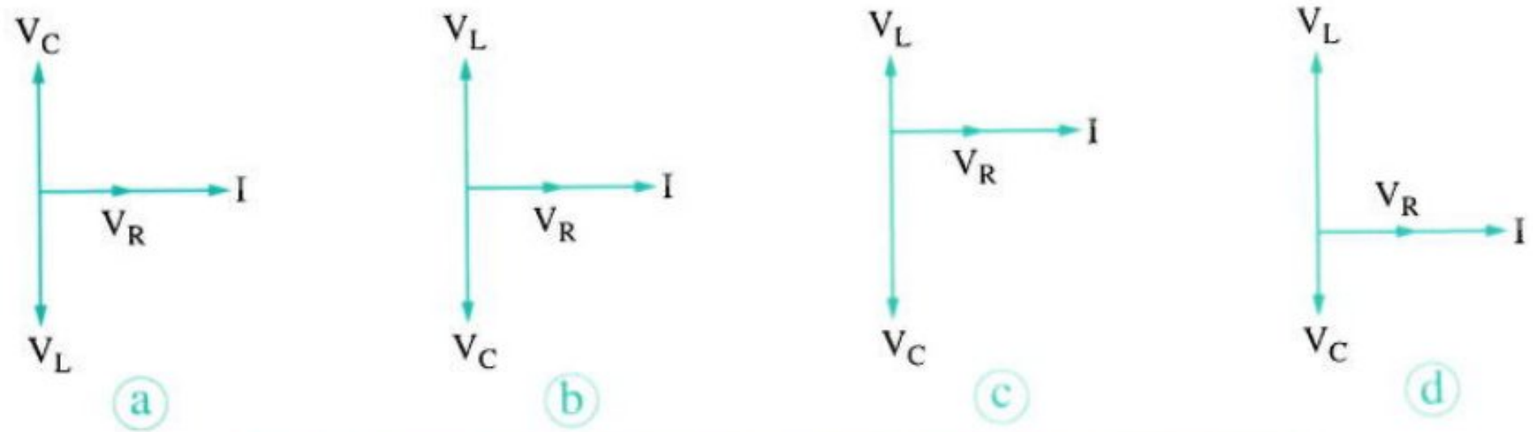
1 When RLC circuit is in a state of resonance, the impedance of the circuit is ..... and equals ..... .

(a) minimum - the ohmic resistance
(b) maximum - the ohmic resistance
(c) minimum - the inductive reactance
(d) maximum - the capacitive reactance

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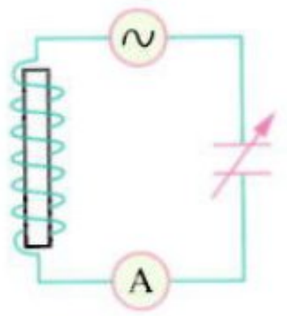
2 The following figures represent phasor diagrams of the voltage and the current, which of the following figures represents the resonance state in RLC circuit?



3 In an AC circuit, it is noticed that by increasing the frequency of the source, the effective value of the current through the circuit increases to a certain value then decreases, so this circuit contains ..... .

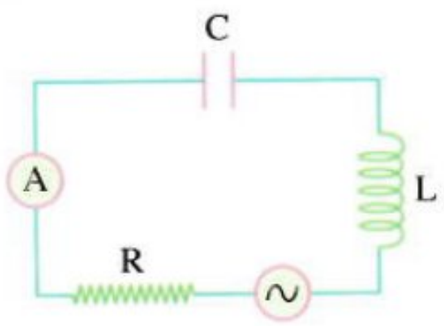
(a) an inductor, a capacitor and an ohmic resistance
(b) an ohmic resistance and an inductor
(c) an ohmic resistance and a capacitor
(d) an ohmic resistance only

4 In the opposite figure, when the iron core is removed from the coil of the RLC circuit that was in a state of resonance, the reading of the hot wire ammeter ..... .



(a) decreases
(b) increases
(c) remains constant
(d) becomes zero

5 The opposite RLC circuit is in a state of resonance and it is connected to an AC source of constant effective value, so by increasing the frequency of the source, then:



(i) The ohmic resistance (R) ..... .

(a) remains constant
(b) decreases to half
(c) increases to the double
(d) increases to three times

(ii) The reading of the hot wire ammeter (A) ..... .

(a) remains constant
(b) decreases because the impedance increases
(c) decreases because the impedance decreases
(d) increases because the impedance increases

6 The resonant frequency in an RLC circuit is determined by the value of ..... .

(a) ohmic resistance and self-inductance of the coil
(b) self-inductance of the coil and capacitance of the capacitor
(c) ohmic resistance and capacitance of the capacitor
(d) ohmic resistance, self-inductance of the coil, and capacitance of the capacitor

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7 In an RLC circuit at resonance, the product of the self-inductance (L) of the coil and the capacitance (C) of the capacitor is ..... .

(a) the angular frequency of current

(b) the square of the angular frequency of current

(c) the reciprocal of the angular frequency of current

(d) the reciprocal of the square of the angular frequency of current

8 To decrease the resonance frequency in the RLC circuit, we could ..... .

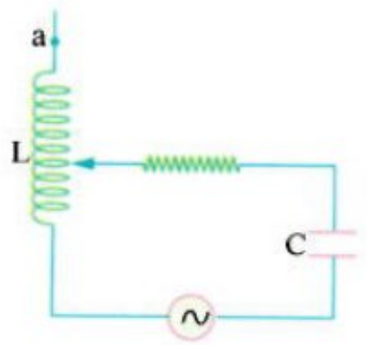
(a) move the turns of the inductor away from each other

(b) cut a part of the coil and reconnect the remaining part in the circuit

(c) remove the capacitor from the circuit (C)

(d) connect an external capacitor in parallel with the capacitor of the circuit

9 In the opposite RLC circuit, what happens to the resonant frequency when moving the slider towards point a?



(a) Increases.

(b) Decreases.

(c) Remains unchanged.

(d) Becomes zero.

10 In a resonance circuit, the capacitance of the capacitor increased to the double and the self inductance of the coil decreased to its eighth, then the frequency of resonance of the circuit would ..... .

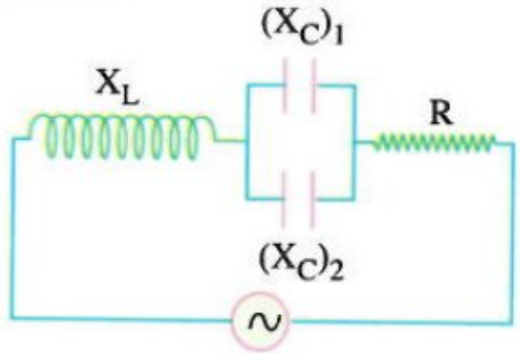
(a) increase to its double

(b) decrease to its half

(c) become four times its original value

(d) become quarter its original value

11 The opposite circuit becomes in a state of resonance when .....



(a) $X_L = (X_C)_1 + (X_C)_2$

(b) $X_L = \frac{(X_C)_1}{2} + \frac{(X_C)_2}{4}$

(c) $X_L = \frac{(X_C)_1 (X_C)_2}{(X_C)_1 + (X_C)_2}$

(d) $X_L = (X_C)_1 = (X_C)_2$

12 A series RLC circuit is connected to an AC source of changeable frequency, so when the frequency of the current becomes less than the resonance frequency of this circuit, then the circuit will have ..... .

(a) capacitive properties because $X_L > X_C$

(b) capacitive properties because $X_L < X_C$

(c) inductive properties because $X_L > X_C$

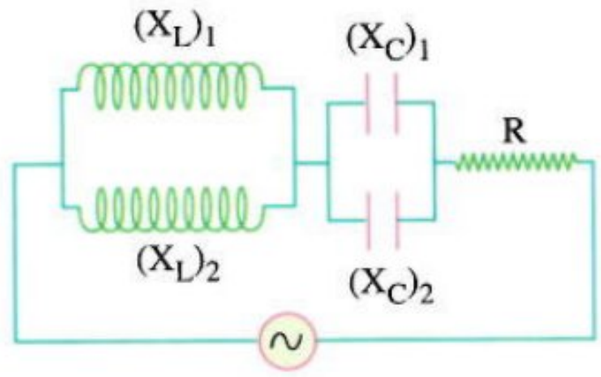
(d) inductive properties because $X_L < X_C$

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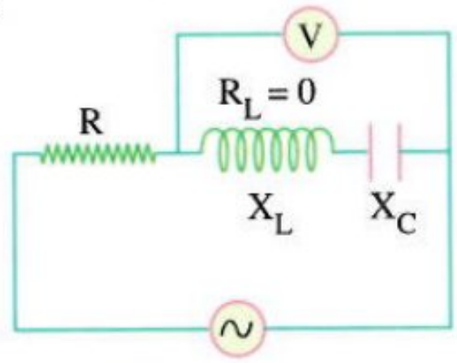
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13 ✱ In the opposite circuit, if $(X_L)_1 = (X_L)_2 = (X_C)_1 = (X_C)_2 = R$, then the circuit has ..... .



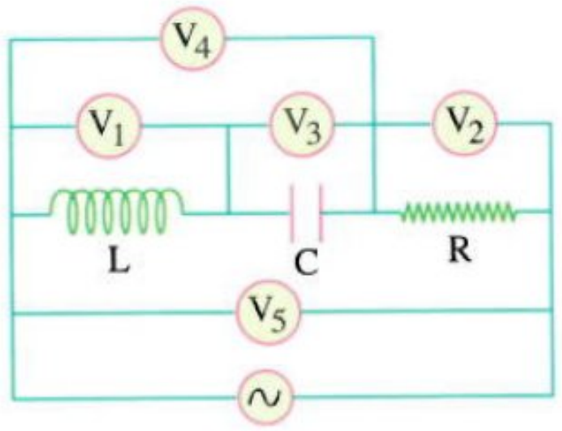
(a) inductive properties
(b) ohmic properties
(c) capacitive properties
(d) inductive or capacitive properties

14 The opposite figure shows an AC circuit. If the consumed power in the resistor (R) is the maximum, then the voltmeter reads .....



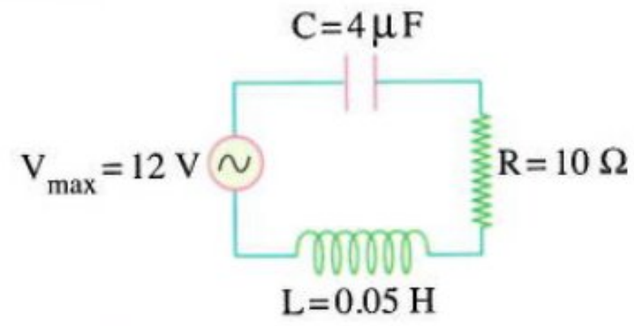
(a) $\sqrt{2V_L^2 - V_C^2}$
(b) $\sqrt{V_R^2 + V_L^2}$
(c) $\sqrt{V_R^2 + (V_L - V_C)^2}$
(d) 0

15 Which of the shown voltmeters in the opposite circuit, has a reading of zero at resonance?



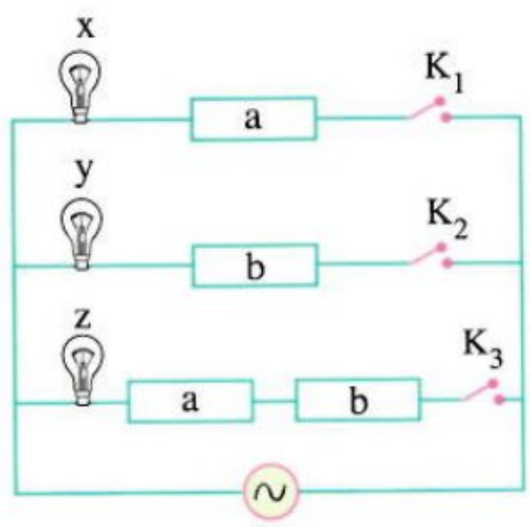
(a) V_1, V_3
(b) V_2
(c) V_4
(d) V_5

16 ✱ The opposite AC circuit is in a state of resonance, so the consumed electric power from the source is ..... .



(a) 0
(b) 3.6 W
(c) 7.2 W
(d) 14.4 W

17 In the opposite electric circuit, the electrical components (a) and (b) are pure, and lamps x, y, and z are identical. After closing switch K_1 only, the total voltage leads the current in phase, and lamp x consumes an electrical power equal to P_w. And after closing switch K_2 only, the total voltage lags behind the current in phase, and lamp y also consumes an electrical power equal to P_w. Then, if only switch K_3 is closed:



(i) lamp z consumes electrical power ..... .

(a) equal to P_w
(b) greater than P_w
(c) less than P_w
(d) whose magnitude cannot be determined

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(ii) the total voltage ..... .

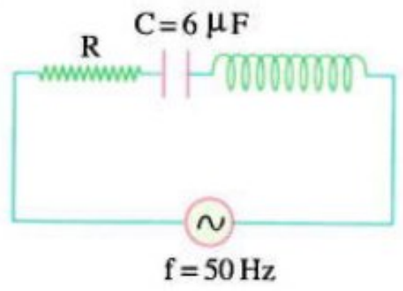
(a) leads the current in phase

(b) lags the current in phase

(c) is in phase with the current

(d) phase cannot be determined relative to the current

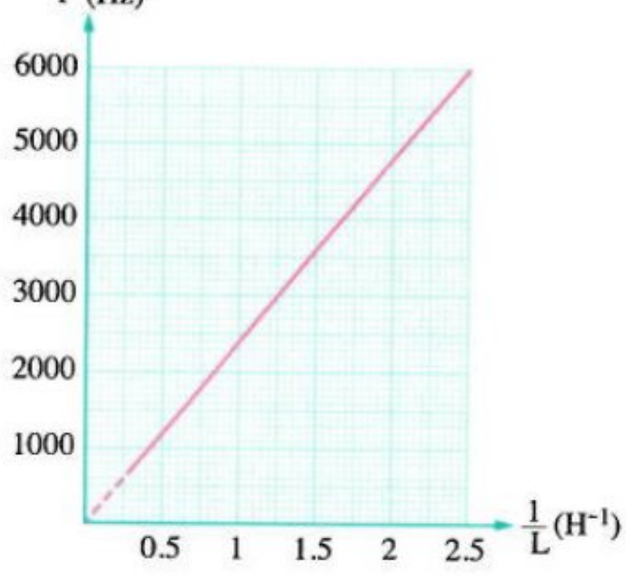
18 In the illustrated circuit, if the impedance of the circuit equals R, then the self inductance of the coil is ..... .



(a) 6 H (b) 1.69 H

(c) 60.731 H (d) 80.41 H

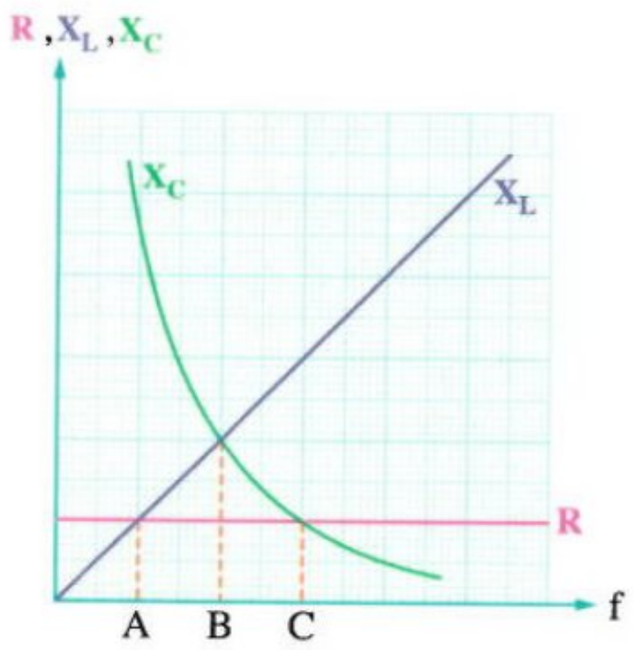
19 A capacitor of constant capacitance is connected in series with an inductor of variable inductance and an AC source. The opposite graph represents the relation between the square of the resonance frequency (f^2) for the circuit and the reciprocal of inductance of the inductor ($\frac{1}{L}$), so the capacitance of the capacitor equals ..... .



(a) 0.85×10^{-5} F (b) 1.05×10^{-5} F

(c) 2.02×10^{-5} F (d) 3.06×10^{-5} F

20 The opposite graph represents the change in the inductive reactance of the pure inductor (X_L), the capacitive reactance of the capacitor (X_C), and the ohmic resistance (R) versus frequency (f) in a series RLC, AC circuit, so:



(i) At which of the following frequencies is the voltage across the resistance equal to the voltage across the inductor?

(a) A (b) B

(c) C (d) A, B or C

(ii) At which of the following frequencies does the circuit have capacitive properties?

(a) A (b) B (c) C (d) A, B or C

(iii) What is the impedance of the circuit at frequency B in terms of the ohmic resistance R in the circuit?

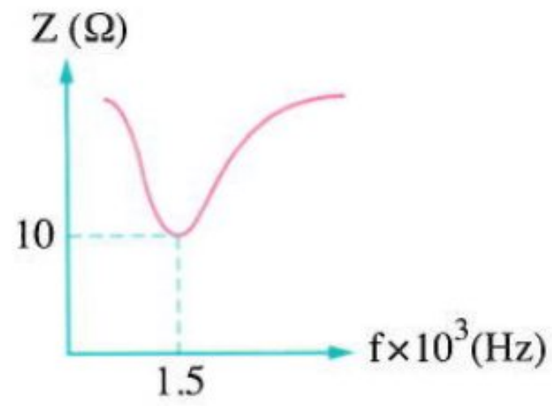
(a) R (b) $\sqrt{2}$ R (c) $\sqrt{3}$ R (d) 2 R

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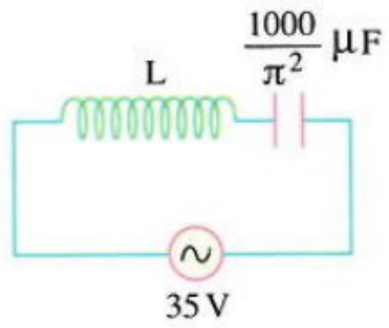
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21 An AC circuit consists of a resistance, a capacitor and an inductor that are connected in series to an AC source of changeable frequency and the opposite graph represents the relation between the impedance (Z) of the circuit and the frequency of the current (f), so the value of the ohmic resistance in this circuit equals ..... .



(a) 1.5 Ω (b) 5 Ω (c) 6.67 Ω (d) 10 Ω

22 * When studying the impedance of the electric circuit shown in the figure while changing the frequency of the AC source, we have obtained the shown graph if the internal resistance of the source is negligible, so :

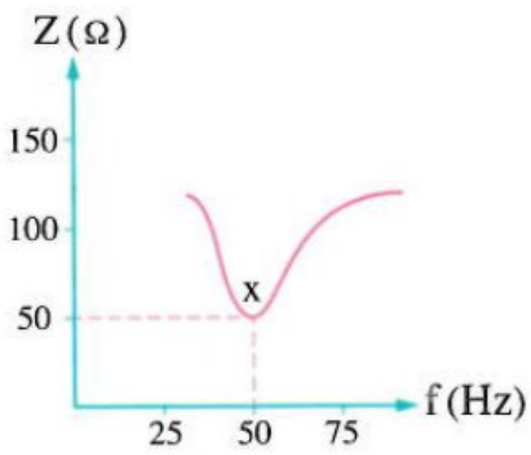


(i) The self inductance of the coil equals ..... .

(a) 0.1 H (b) 0.2 H

(c) 2 H (d) 1 H

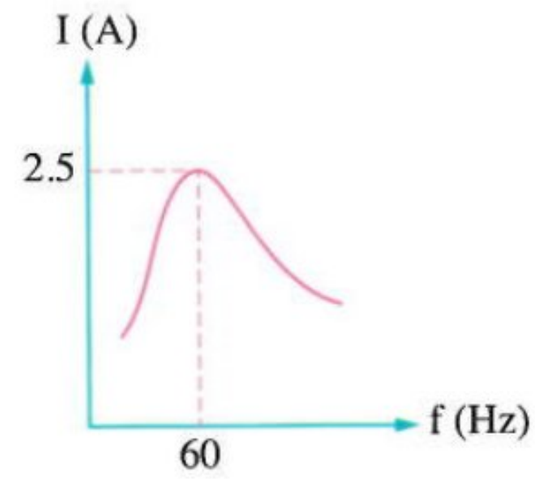
(ii) The potential differences between the terminals of each of the coil and the capacitor at position x respectively are ..... , ...



(a) 22 V, 22 V (b) 41.34 V, 22 V

(c) 41.34 V, 41.34 V (d) 22 V, 41.34 V

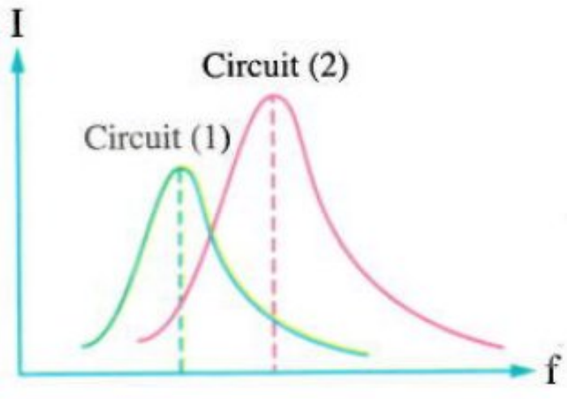
23 The opposite figure represents the graphical relation between the effective value of the current (I) that passes in an RLC circuit and the frequency of the source (f). If the capacitance of the capacitor is 2.58×10^{-4} F, then the self inductance of the coil that makes the circuit in a state of resonance is approximately equal to ..... .



(a) 15 mH (b) 22 mH

(c) 27 mH (d) 32 mH

24 Two RLC, AC circuits. The electrical source in each has a constant voltage (V) while its frequency can be changed. And the opposite graph represents the relationship between the value of the current (I) flowing in each of the two circuits and the frequency (f) of the source. If the values of the equivalent capacitances in the two circuits are equal, which of the two circuits:



(i) has a greater ohmic resistance?

(a) Circuit (1). (b) Circuit (2).

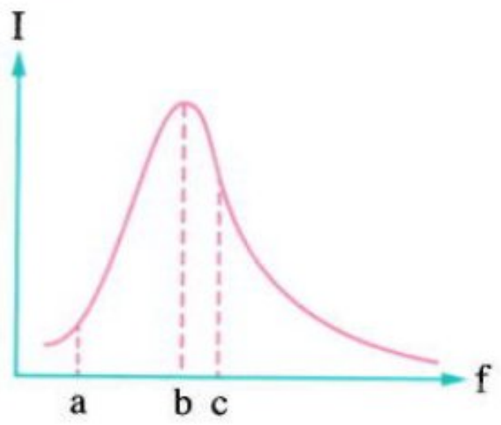
(c) Equal for both circuits. (d) The answer cannot be determined.

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(ii) has a greater total self-inductance?

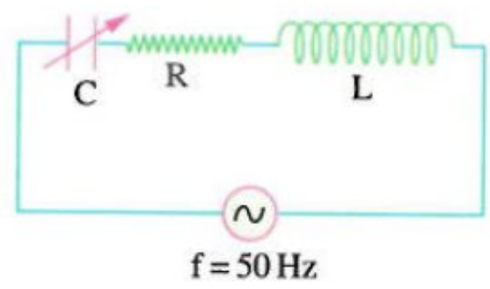
(a) Circuit (1).
(b) Circuit (2).
(c) Equal for both circuit.
(d) The answer cannot be determined.

25 An RLC, AC circuit with components connected in series in which electrical source's frequency can be changed while the effective value of its voltage remains constant. And the opposite graph represents the relationship between the value of the current (I) flowing in the circuit and the frequency (f) of the source, so the frequency at which the value of phase angle between the total voltage and the current is ..... .



	Larger	Smaller
(a)	a	b
(b)	a	c
(c)	b	c
(d)	c	b

26 In the illustrated figure:
If the circuit is in a state of resonance, then the capacitance of the capacitor gets increased, to its double, so the new frequency that keeps the state of resonance is ..... .



(a) 500 Hz
(b) $25\sqrt{2}$ Hz
(c) 50 Hz
(d) $100\sqrt{2}$ Hz

27 A series RLC, AC circuit in which the electrical source frequency can be changed while the effective value of its voltage remains constant. Figure (1) graphically represents the relationship between the impedance (Z) of the circuit and the frequency (f) of the source. Figure (2) graphically represents the relationship between the value of the current (I) flowing through the circuit and the frequency (f) of the source.
If the ohmic resistance of the inductor is negligible and the capacitance of the capacitor is $(\frac{1}{\pi}\ \mu F)$, so:

(i) The potential difference across the inductor at a frequency of 500 Hz equals ..... .

(a) 1 kV
(b) 2 kV
(c) 3 kV
(d) 4 kV

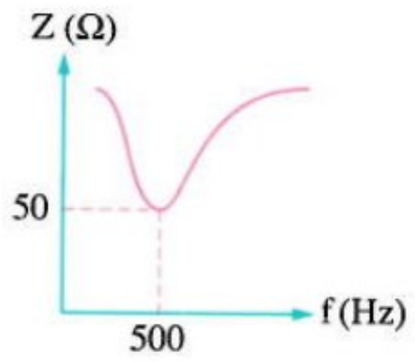


Figure (1)

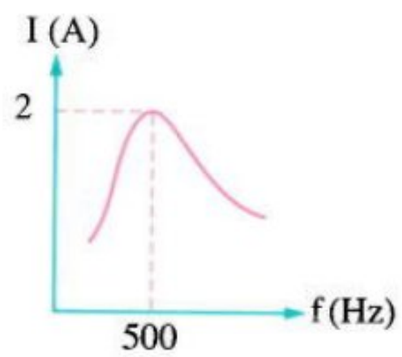


Figure (2)

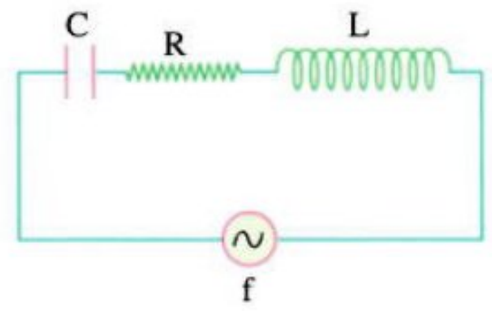
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(ii) The potential difference across the capacitor at a frequency of 250 Hz is approximately equal to ..... .

(a) 33 V (b) 66 V (c) 133 V (d) 166 V

28 In the circuit illustrated in the opposite figure:

Which of the following configurations causes a resonance state?



(a)

f	C	L
1000 Hz	1 μF	1 H

(b)

f	C	L
100 Hz	10 μF	10 H

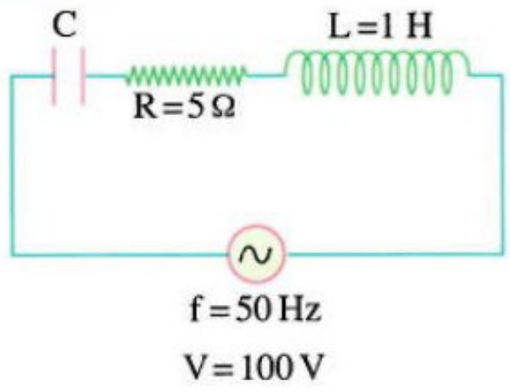
(c)

f	C	L
400 Hz	2 μF	2 H

(d)

f	C	L
500 Hz	$\frac{7}{22}$ μF	$\frac{7}{22}$ H

29 In the circuit illustrated in the opposite figure, if the current passing through it is 20 A, then:



(i) The capacitance of capacitor C is ..... .

(a) 10^{-5} μF (b) 5 F

(c) 10^{-5} F (d) 98596 F

(ii) The potential difference across the coil is ..... .

(a) 0 (b) 50 V (c) 100 V (d) 6285.7 V

30 * A resonance circuit containing a capacitor of capacitance 30 μF receives a wave of frequency 750 kHz, if the coil (L_1) is replaced with another of self inductance (L_2) that is 5 times as that of the first coil and the capacitance of the capacitor is increased by 32 μF, then: (Where: the speed of the electromagnetic waves = 3×10^8 m/s)

(i) The frequency of the wave that will be received equals ..... .

(a) 7.3×10^4 Hz (b) 2.33×10^5 Hz (c) 4.8×10^5 Hz (d) 24.1×10^5 Hz

(ii) The wavelength of the wave that can be received equals ..... .

(a) 6.99×10^{13} m (b) 2.25×10^{13} m (c) 1.29×10^3 m (d) 4×10^2 m

(iii) The inductances of the coil in the two cases L_1, L_2 respectively are ..... , ...

(a) 7.1×10^{-3} H, 3.5×10^{-2} H (b) 1.12×10^{-3} H, 7.1×10^{-3} H

(c) 1.5×10^{-9} H, 7.5×10^{-9} H (d) 7.5×10^{-9} H, 1.5×10^{-9} H

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31 ✱ A tuning circuit in a radio receiver consists of an inductor of inductance 10 mH, a resistance 50 Ω and a capacitor of variable capacitance. When wireless waves of frequency 980 kHz are received by the antenna, a voltage 10^{-4} V is generated across the circuit, then:

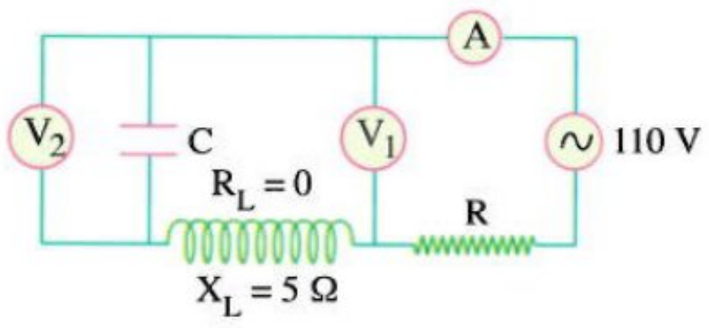
(i) The capacitance of the capacitor which is required to fulfill the resonance state equals ..... .

(a) 8.3×10^{-12} F (b) 2.6×10^{-12} F (c) 8.4×10^{-5} F (d) 1.6×10^{-5} F

(ii) The value of the current in the circuit in the state of resonance equals ..... .

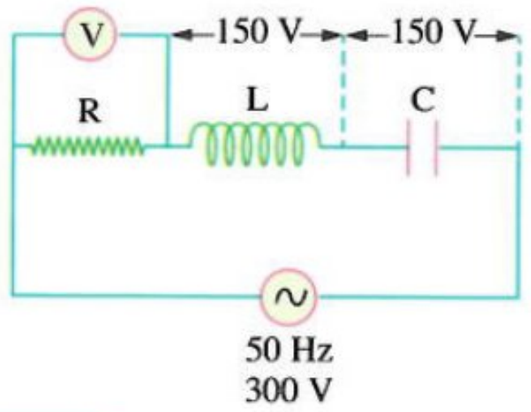
(a) 1.6×10^{-9} A (b) 2×10^{-7} A (c) 2×10^{-6} A (d) 3×10^{-6} A

32 In the shown AC circuit, if the reading of the ammeter is 2 A and the reading of voltmeter V_1 equals zero, the value of resistance R and the reading of voltmeter V_2 respectively are ..... .



(a) 45 Ω, 5 V (b) 50 Ω, 8 V (c) 55 Ω, 10 V (d) 60 Ω, 20 V

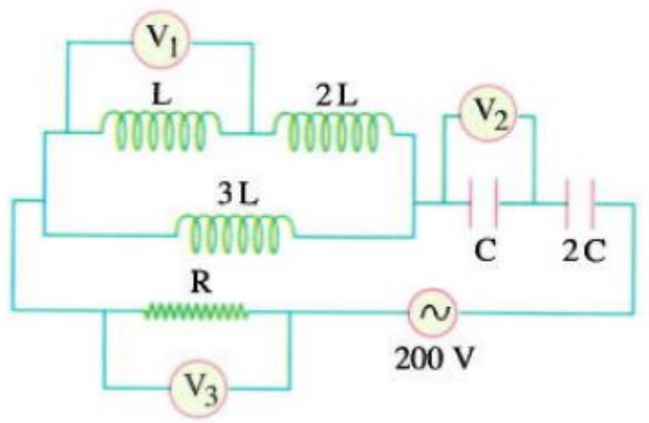
33 In the opposite circuit, the reading of the voltmeter is ..... . (Where the ohmic resistances of the source and the coil are negligible)



(a) zero (b) 100 V

(c) 200 V (d) 300 V

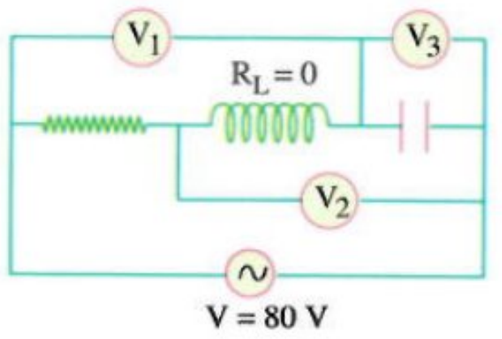
34 In the opposite electric circuit, the inductors have negligible ohmic resistance, and the readings of the two voltmeters V_1 and V_2 are 50 V and 100 V respectively, so the reading of voltmeter V_3 equals ..... .



(a) 50 V (b) 100 V

(c) 150 V (d) 200 V

35 ✱ In the opposite electric circuit, the readings of the two voltmeters V_1 and V_2 are 100 V and 0 V respectively, so the reading of voltmeter V_3 equals ..... .



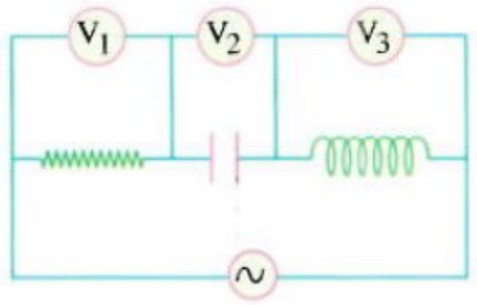
(a) 0 V (b) 60 V

(c) 80 V (d) 100 V

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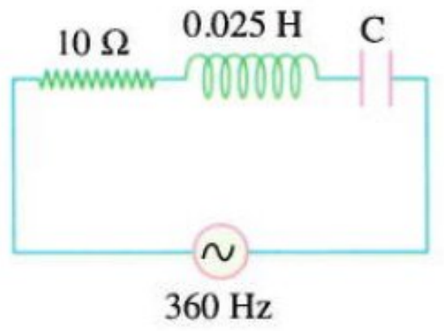
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36 ✱ In the opposite figure, a series RLC, AC circuit is in a state of resonance, and the value of the current flowing through it is 1 A. If the readings of voltmeters V_1, V_2, and V_3 respectively are 20 V, 60 V, 100 V, the total ohmic resistance in the circuit equals ..... .



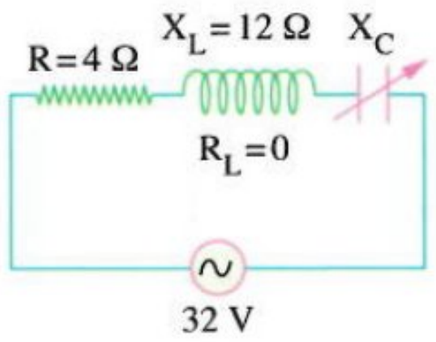
(a) 20 Ω (b) 60 Ω (c) 80 Ω (d) 100 Ω

37 In the opposite electric circuit, to make the current and the total voltage be in phase, the capacitance of the capacitor must be ..... .



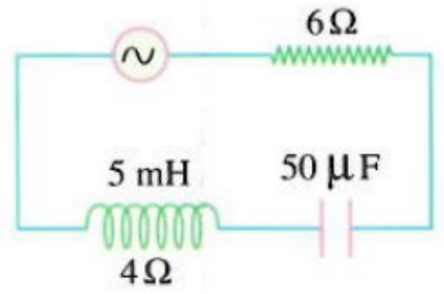
(a) 2.8×10^{-3} F (b) 5.6×10^{-3} F
(c) 7.8 μF (d) 3.9 μF

38 The opposite diagram represents an RLC circuit that contains a capacitor of changeable capacitance, so the highest effective value of current that can pass through the circuit equals ..... .



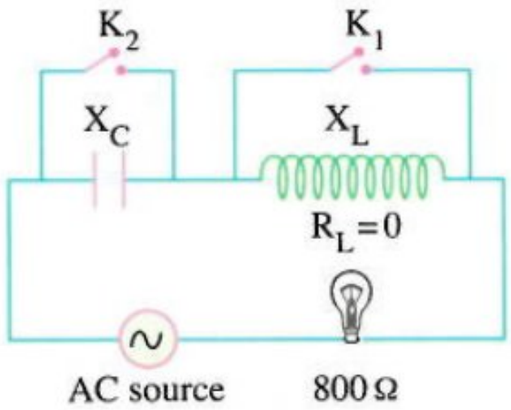
(a) 2 A (b) 4 A
(c) 6 A (d) 8 A

39 ✱ The opposite figure shows an AC circuit, the voltage of its source is given by the relation; V = 20 sin (ωt) where ω = 2000 rad/s, so the maximum value of the current that passes through the circuit equals ..... .



(a) 2 A (b) 3.3 A
(c) $2\sqrt{5}$ A (d) $\sqrt{5}$ A

40 ✱ In the opposite circuit, AC source of frequency 50 Hz and emf 220 V is connected to a capacitor of capacitance 4 μF and an inductor of self inductance $\frac{1225}{484}$ H, then:



(i) The impedance of the circuit if the two switches K_1, K_2 are open is ..... .

(a) 800 Ω (b) 765.45 Ω
(c) 150.6 Ω (d) 0

(ii) The ratio between the impedance of the circuit at closing the switch (K_1) only to its impedance at closing the switch (K_2) only equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{1}{2}$ (c) $\frac{2}{1}$ (d) $\frac{1}{3}$

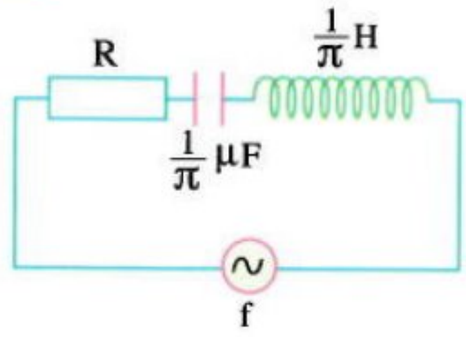
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(iii) At closing the two switches K_1, K_2, ...

	The lamp lighting	The impedance
(a)	increases	795.45 Ω
(b)	increases	800 Ω
(c)	remains constant	795.45 Ω
(d)	remains constant	800 Ω

41 In the opposite figure:

The circuit shows an AC source of variable frequency (f) and constant voltage, so the voltage across resistance R reaches maximum value at frequency ..... .



(a) 0 (b) 100 Hz

(c) 250 Hz (d) 500 Hz

42 The two figures (1), (2) show two parts of two AC circuits, if the resonance frequency in figure (1) equals 10 kHz, so the resonance frequency in figure (2) equals ..... .

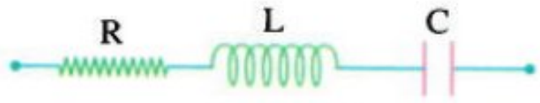


Figure (1)

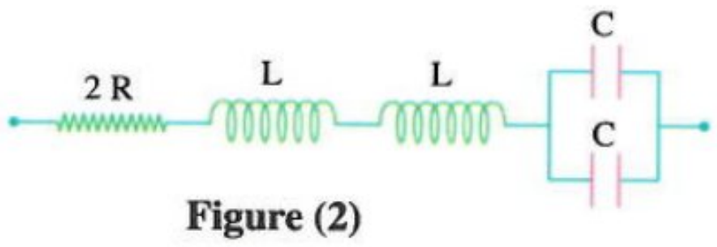
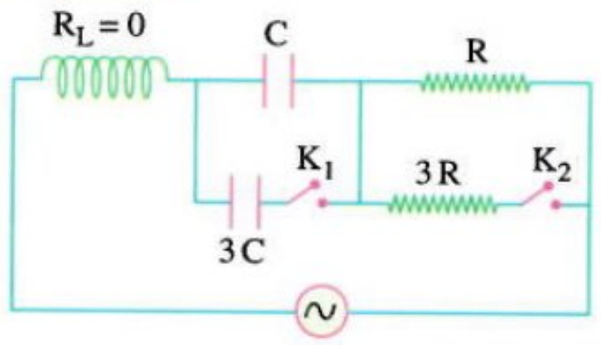


Figure (2)

(a) 2.5 kHz (b) 5 kHz

(c) 10 kHz (d) 40 kHz

43 In the opposite electrical circuit, if the frequency that achieves the resonance state when switches K_1 and K_2 are open is f_o, then the frequency that achieves the resonance state after:



(i) closing switch K_1 only is ..... .

(a) $\frac{1}{4} f_o$ (b) $\frac{1}{2} f_o$ (c) f_o (d) $2 f_o$

(ii) closing switch K_2 only is ..... .

(a) $\frac{1}{4} f_o$ (b) $\frac{1}{2} f_o$ (c) f_o (d) $2 f_o$

44 * A wireless transmission circuit contains a resonant circuit which consists of an inductor of self inductance $\frac{49}{121}$ mH and a capacitor where the potential difference between its plates becomes 9 V when one of its plates is loaded with a charge of 36 mC, then:

(i) The frequency of the resonant circuit is ..... .

(a) 25 Hz (b) 41.67 Hz (c) 62.5 Hz (d) 125 Hz

(ii) The reactances of each of the coil and the capacitor respectively are ..... , ...

(a) $\frac{7}{22}$ Ω, $\frac{22}{7}$ Ω (b) $\frac{22}{7}$ Ω, $\frac{7}{22}$ Ω (c) $\frac{22}{7}$ Ω, $\frac{22}{7}$ Ω (d) $\frac{7}{22}$ Ω, $\frac{7}{22}$ Ω

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45 ✱ AC circuit consists of a source of frequency 50 Hz, a capacitor of capacitance $\frac{700}{22}$ μF, an ohmic resistance 50 Ω and an inductor with a negligible ohmic resistance where all are connected in series. When the potential difference between parts of the circuit is measured, it's found that the potential difference across the capacitor is equal to that across the inductor = 20 V, then:

(i) The inductance of the inductor equals ..... .

(a) $\frac{22}{7}$ H (b) $\frac{7}{44}$ H (c) $\frac{7}{22}$ H (d) $\frac{44}{7}$ H

(ii) The maximum value of the emf of the source is ..... .

(a) $10\sqrt{2}$ V (b) 10 V (c) $5\sqrt{2}$ V (d) 5 V

46 ✱ A battery of emf 12 V is connected in series with an inductor, so the current passing in the circuit was 2 A. When the battery is replaced by an AC source that has an effective value of voltage 12 V, the effective value of the current passing in the circuit became 1.2 A and when a capacitor is connected in series with the inductor in the second circuit the electric current intensity returned to its original value as that of the first circuit (Ignoring the internal resistance of the two voltage sources), then:

(i) The ohmic resistance of the inductor equals ..... .

(a) 4 Ω (b) 6 Ω (c) 8 Ω (d) 10 Ω

(ii) The inductive reactance of the inductor equals ..... .

(a) 4 Ω (b) 6 Ω (c) 8 Ω (d) 10 Ω

(iii) The last circuit that contains an AC source, an inductor and a capacitor ..... .

(a) has capacitive properties (b) is in a state of resonance

(c) has inductive properties (d) has capacitive and inductive properties

47 ✱ An inductor of self inductance 0.08 H and of resistance 30 Ω is connected to an AC source of 10 V and frequency 80 Hz, then to make the phase angle between the total voltage and the current decrease to zero without changing the current intensity in the coil while keeping the same AC source, what should be added to the circuit ?

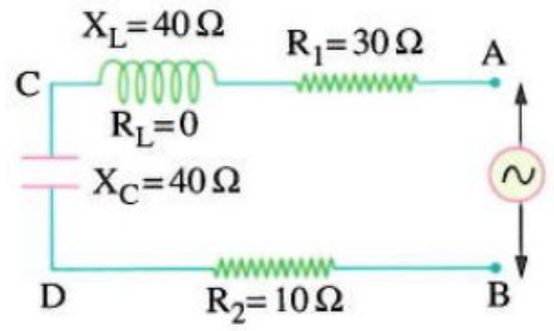
(a) A capacitor of capacitive reactance 6 Ω.

(b) A capacitor of capacitive reactance 8 Ω.

(c) A resistance of 20.18 Ω and a capacitor of capacitance 100 μF.

(d) A resistance of 20.18 Ω and a capacitor of capacitance 49.49 μF.

48 ✱ In the opposite figure, the points A and B are connected to an AC power source of emf 200 volts and frequency 50 Hz, then:



(i) The potential difference between A and C equals ..... .

(a) 350 V (b) 250 V

(c) 140 V (d) 100 V

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(ii) The potential difference between B and C equals ..... .

(a) 137.44 V (b) 155.2 V (c) 206.16 V (d) 250 V

(iii) The consumed power in the circuit equals ..... .

(a) 250 W (b) 600 W (c) 750 W (d) 1000 W

49 ✳ AC source is connected in series in a circuit containing an inductor with a negligible resistance, a capacitor and an ohmic resistor of 100 Ω, thus a maximum current flows and when the source is replaced with another having the same emf and double the frequency of the first, the electric current intensity decreased to 0.45 of its original intensity, then each of the capacitive and inductive reactances in the first case respectively are ..... , ...

(a) 79.38 Ω, 79.38 Ω (b) 79.38 Ω, 132.3 Ω (c) 132.3 Ω, 132.3 Ω (d) 132.3 Ω, 79.38 Ω

50 ✳ If the impedance of an RLC circuit in the case of resonance is 8 Ω and its frequency 60 Hz and when the frequency changes to 80 Hz, the impedance of the circuit becomes 10 Ω, then:

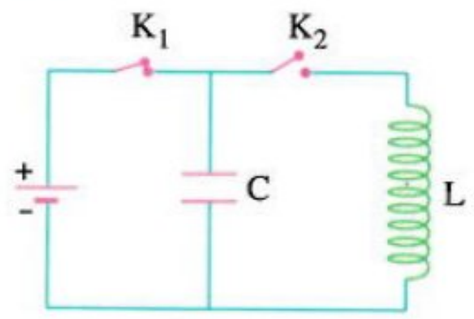
	The self inductance of the coil	The capacitance of the capacitor
(a)	0.027 H	2.58×10^{-4} F
(b)	0.015 H	2.58×10^{-4} F
(c)	0.027 H	4.54×10^{-4} F
(d)	0.015 H	6.36×10^{-4} F

51 What change should be done to the self-inductance of the coil in an oscillating circuit to double the frequency of the current?

(a) Decrease it to its quarter. (b) Increase it to its quadruple.

(c) Decrease it to its half. (d) Increase it to its double.

52 In the oscillating circuit, when the switch K_1 is opened and switch K_2 is closed for a while, assuming that the ohmic resistance of the coil and the wires is negligible, which of the following happens in the coil?



(a) DC current passes through it.

(b) Unidirectional current passes through it.

(c) AC current passes through it.

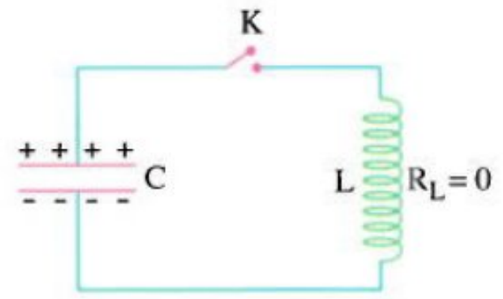
(d) No electric current passes through it.

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53 The opposite figure represents an oscillating circuit consisting of a capacitor with a capacitance of 25.3 μF and an inductor with a self-inductance of 0.4 H. How many times during one second from the moment of closing the switch (K) will:



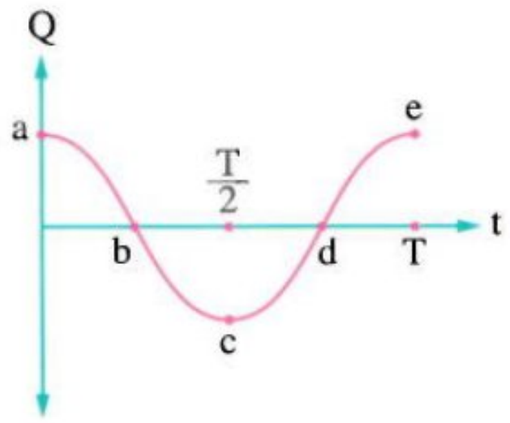
(i) the capacitor be completely discharged of electric charge?

(a) 50 (b) 100 (c) 150 (d) 200

(ii) the magnetic energy in the inductor be equal to the electric energy in the capacitor?

(a) 50 (b) 100 (c) 150 (d) 200

54 The opposite graph represents the relation between the quantity of electric charge (Q) on one of the plates of a capacitor in an oscillating circuit and time (t).



At which of the points shown on the graph:

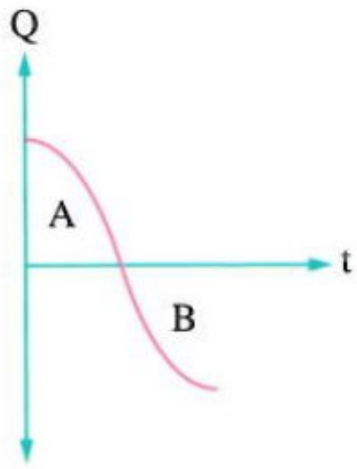
(i) The current intensity through the circuit has the maximum value?

(a) a, e (b) b, d (c) c, e (d) a, c

(ii) The capacitor is charged with the same quantity of charge and has the same polarity?

(a) a, e (b) b, d (c) c, e (d) a, c

55 The opposite graph represents half a cycle of the accumulated charge on a capacitor in an oscillating circuit from the moment at which the circuit switch is closed, so in which of the two stages A or B:



(i) does the value of the potential difference across the capacitor increase?

(a) Stage A (b) Stage B

(c) Both stages. (d) Neither stage.

(ii) does the coil generate a forward induced emf?

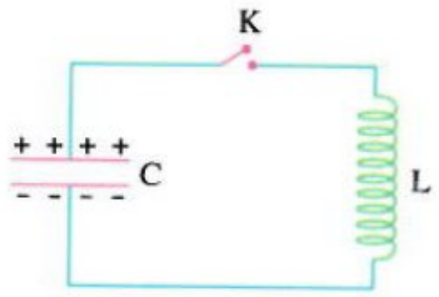
(a) Stage A (b) Stage B (c) Both stages. (d) Neither stage.

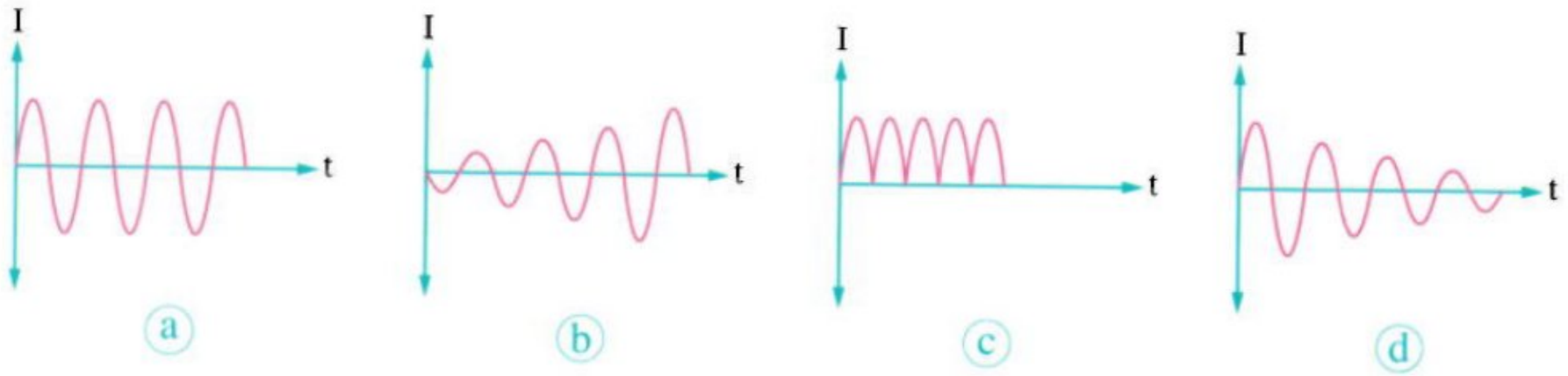
(iii) does the energy stored in the coil increase?

(a) Stage A (b) Stage B (c) Both stages. (d) Neither stage.

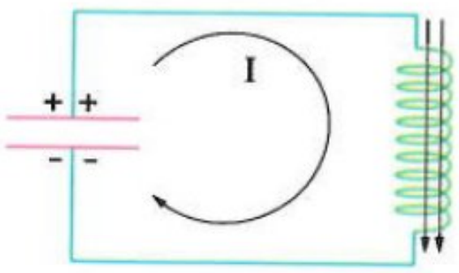
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56 The shown electric circuit contains a charged capacitor C connected to an inductor L with ohmic resistance R through an opened switch K, so the current flowing through the inductor when the switch (K) is closed will be represented by graph ..... .



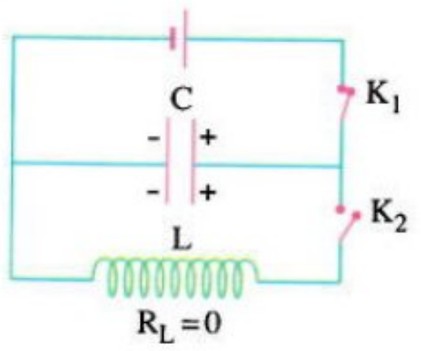


57 The opposite figure represents the direction of the current in the oscillating circuit at a certain moment, what happens to the current intensity (I) in the following interval through the quarter of the periodic time of the current?



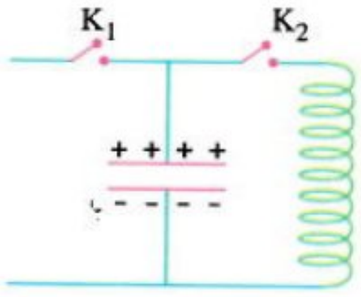
(a) Increases.
(b) Increases then decreases.
(c) Decreases.
(d) Decreases then increases.

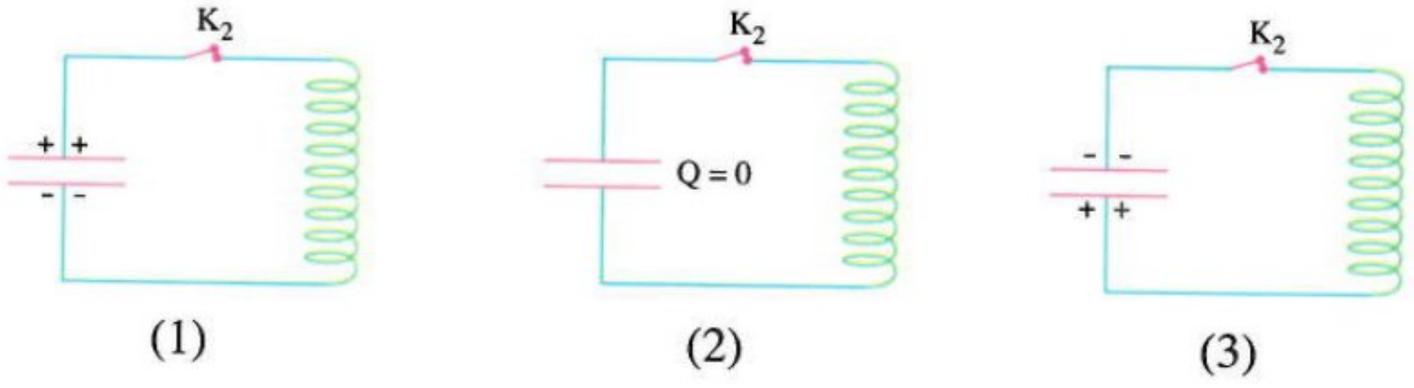
58 The opposite electrical circuit contains an inductor with a self-inductance of 0.25 mH and a capacitor with a capacitance of 1 μF that is carrying a charge Q. If switch K_1 is opened, the capacitor will completely discharge for the first time after closing switch K_2 within a time of ..... .



(a) 10 μs
(b) 20 μs
(c) 25 μs
(d) 100 μs

59 The opposite figure illustrates an oscillating circuit with a charged capacitor. The following figures represent three instants of the circuit during half a cycle of the current after closing switch K_2:





(1) (2) (3)

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So, which of these figures represents:

(i) The transformation of electrical energy into magnetic energy in the coil?

(a) Figure (1) only. (b) Figure (2) only.

(c) Figure (3) only. (d) Both figures (1) and (3).

(ii) The transformation of magnetic energy into electrical energy in the capacitor?

(a) Figure (1) only. (b) Figure (2) only.

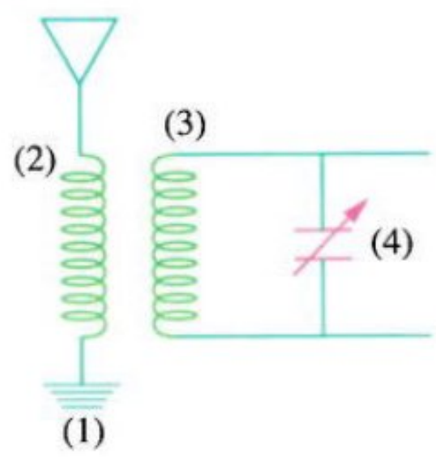
(c) Figure (3) only. (d) Both figures (1) and (3).

(iii) Charging the capacitor?

(a) Figure (1) only. (b) Figure (2) only.

(c) Figure (3) only. (d) Both figures (1) and (3).

60 The opposite figure represents a circuit of a wireless receiver. Which of the following components can be used to control the picked-up broadcast signal?



(a) Component (1). (b) Component (2).

(c) Component (3). (d) Component (4).

61 A tuning (resonant) wireless circuit receives a radio station of frequency f, what is the change required to be done on the tuning circuit to receive a radio wave of frequency 2 f?

(a) Increasing the inductance of the coil to the double and increasing the capacitance of the capacitor to the double.

(b) Increasing the inductance of the coil to the double while decreasing the capacitance of the capacitor to the half.

(c) Decreasing the inductance of the coil to the half while increasing the capacitance of the capacitor to the double.

(d) Decreasing the inductance of the coil to the half and decreasing the capacitance of the capacitor to the half.

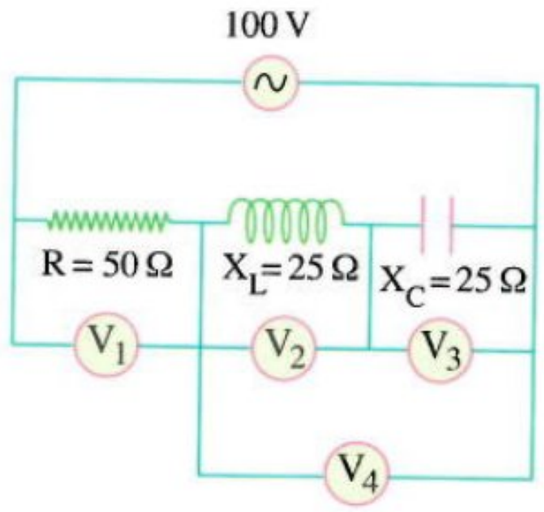
62 The ratio between the impedance of a receiving circuit when it receives a wireless signal of frequency f and its impedance when it receives another wireless signal of frequency 4 f is ..... .

(a) 0.25 (b) 0.5 (c) 1 (d) 4

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Second Essay questions

1 Using the electric circuit illustrated in the figure and the given data, **calculate:**



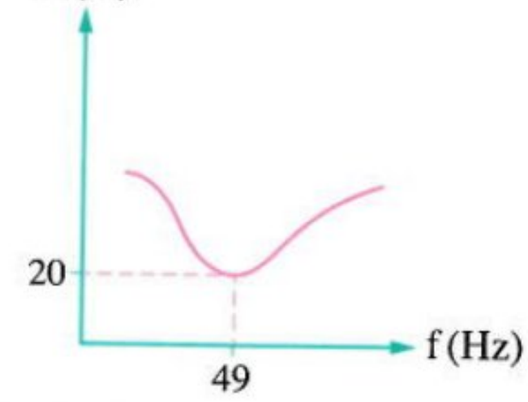
(a) The ratio between the readings of the two voltmeters $\left(\frac{V_1}{V_2}\right)$.

(b) The ratio between the readings of the two voltmeters $\left(\frac{V_2}{V_3}\right)$.

(c) The reading of voltmeter V_4

(d) The ratio $\left(\frac{V_1}{V_{source}}\right)$.

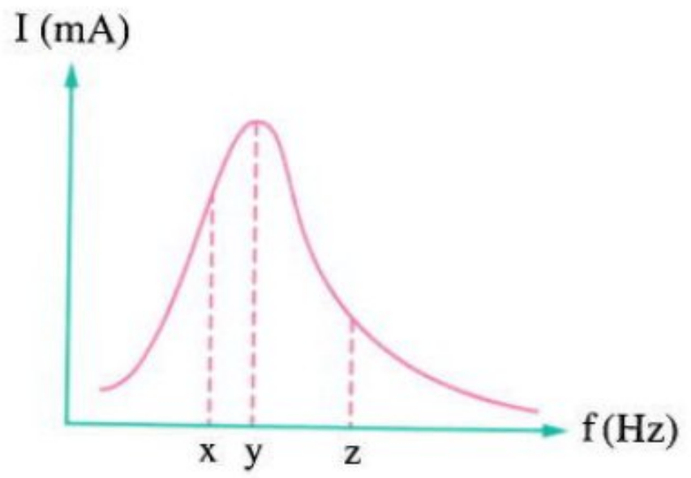
2 The opposite graph represents the relation between the impedance of RLC circuit (Z) and the frequency of AC source (f), so if the capacitance of the capacitor is 1.05×10^{-4} F, **find:**



(a) The ohmic resistance for the circuit.

(b) The self inductance of the coil.

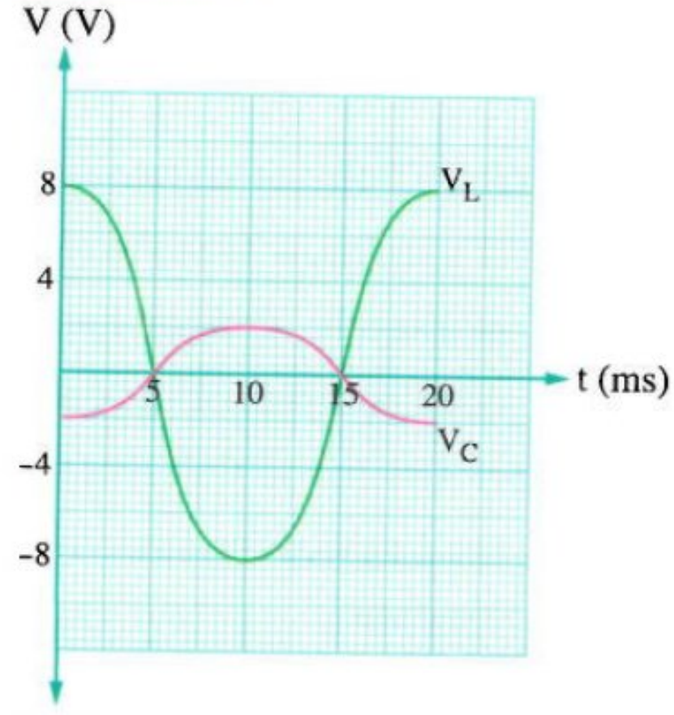
3 The opposite graph represents the relation between the value of the current (I) in an RLC circuit and the frequency of the current (f). **Explain at which of the frequencies x, y, z in the shown graph:**



(a) The inductive reactance is less than the capacitive reactance.

(b) The phase angle between the total voltage and current in the circuit equals zero.

4 The opposite graph represents a sine curve for AC voltage across each of an inductor V_L and a capacitor V_C in RLC circuit:



(a) **Explain why** this circuit isn't in a state of resonance.

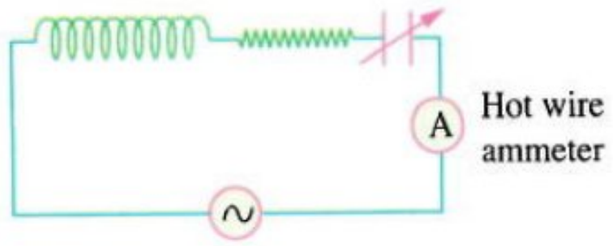
(b) **Calculate** the frequency that makes the circuit in a resonance state.

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5 In the opposite figure, if the circuit is in a state of resonance, **then what happens to the hot wire ammeter reading when increasing:**



(a) the capacitance of the capacitor?

(b) the value of the resistance R?

(c) the frequency of the source at constant voltage?

(d) the voltage of the source at constant current frequency?

---

6 A solenoid coil was connected to a DC source of emf 12 V, so a current of intensity 1 A flew in the circuit and when this source is replaced with an AC source whose effective value of voltage equals that of the DC source and its frequency is 50 Hz, the intensity of the current passing in the circuit became 0.6 A and when a capacitor is connected in series to the coil in this circuit, the electric current intensity returned to its previous value 1 A of the DC circuit. (Ignoring the internal resistance of the two sources), **then find:**

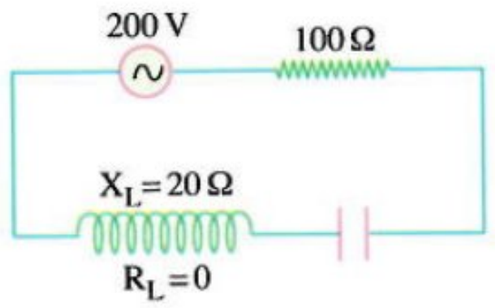
(a) The self inductance of the coil.

(b) The capacitance of the capacitor.

(c) The phase angle between the current and the total voltage in the AC circuit.

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7 In the shown circuit, if the circuit is in resonance and the maximum charge (Q_{max}) that is accumulated on the capacitor is 200 μC, **calculate :**

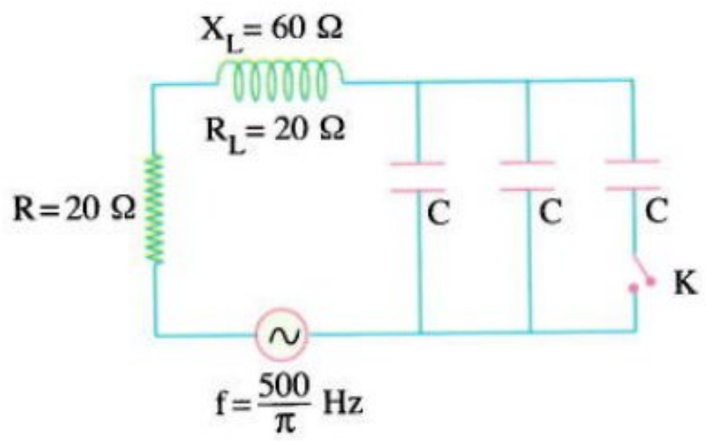


(a) The capacitance of the capacitor.

(b) The resonance frequency.

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8 If the opposite AC circuit is in a state of resonance and the switch K is opened. **Calculate:**

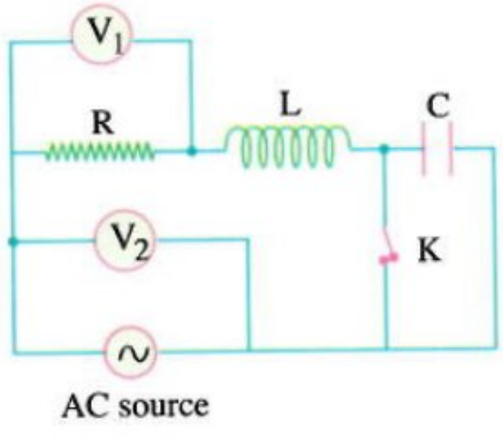


(a) The capacitance of capacitor C

(b) The phase angle between the total voltage and the current at closing the switch K.

---

9 In the AC circuit shown in the figure:



(a) If you know that the ohmic resistance of the coil L is negligible, **deduce** the potential difference across its ends in terms of the two voltmeters readings V_1, V_2

(b) It was found that when the switch K was opened, the readings of the two voltmeters V_1, V_2 became equal, **explain.**

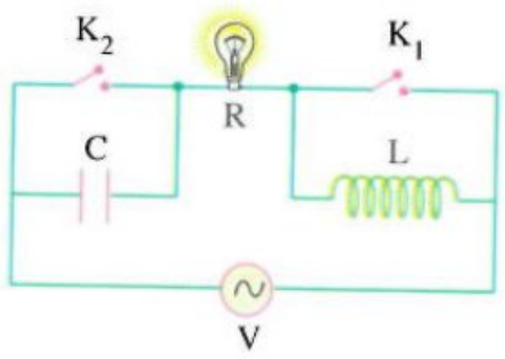
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10 RLC circuit in a resonance state as shown in the opposite figure, **explain what will happen to the brightness of the bulb in each of the following cases:**

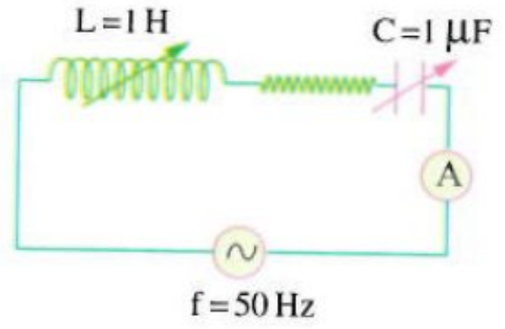
(a) closing the switch K_1 only

(b) closing the switch K_2 only

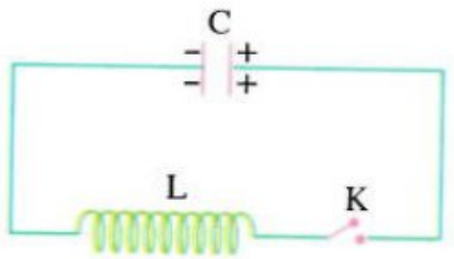
(c) closing the two switches K_1, K_2 together.



11 The opposite figure represents AC circuit containing a dynamo of frequency 50 Hz, an inductor of self inductance 1 H, a capacitor of capacitance 1 μF and hot wire ammeter. **Suggest** three different methods by which the ammeter reading can be increased to its maximum possible value without changing the value of the ohmic resistance in the circuit.



12 The opposite figure shows a capacitor C of charged plates with charge Q being connected to an inductor L of negligible ohmic resistance by using a switch K, **explain why** the two plates of the capacitor exchange charges when the switch K is closed.

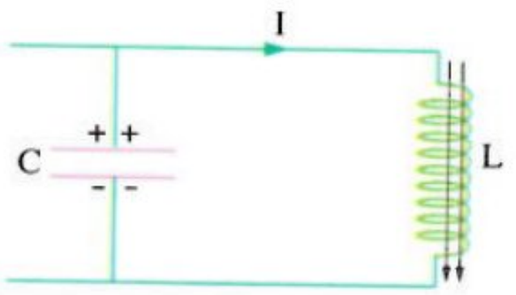


13 The opposite figure shows an oscillating circuit at a certain instant. At that moment:

(a) **Is** the capacitor charging or discharging?

(b) **Does** the magnetic energy inside the coil increase or decrease? **Explain.**

(c) **Explain why** the process of charging and discharging the capacitor does not continue indefinitely.



14 **Explain how** a radio receiver can pick up a certain radio wave.

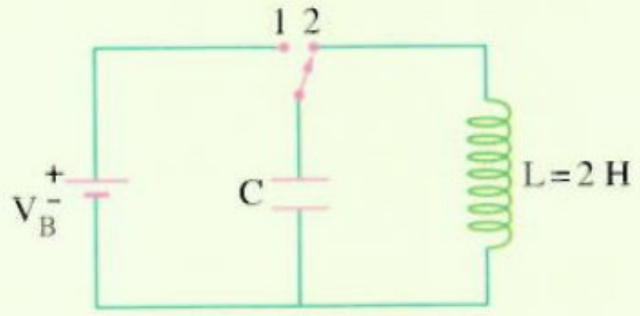
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Questions of the Previous Years Exams on Lesson 3

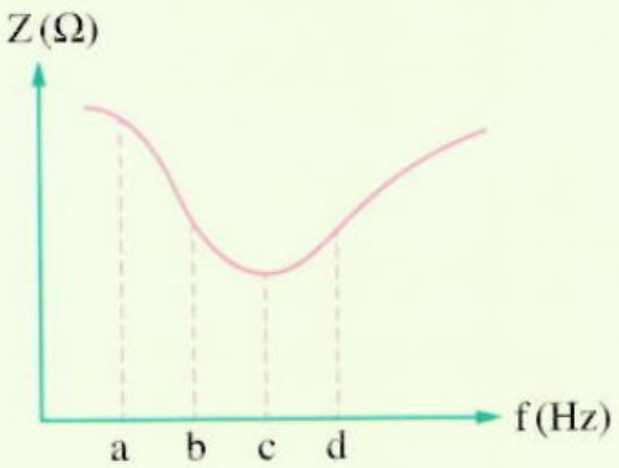
1 The shown oscillatory circuit contains an inductor of inductance (L = 2 H), so the capacitance of capacitor (C) that is required to obtain a current of frequency 80 Hz equals ..... · (Take: $\pi = 3.14$) *(Exp. June 21, Practice 25)*



(a) 1.98 μF (b) 1.98×10^{-6} μF

(c) 1.58×10^{-4} μF (d) 1.58 μF

2 An alternating current circuit contains an inductive coil of negligible ohmic resistance, a capacitor of variable capacitance and an ohmic resistance in series. From the figure, the source voltage becomes equal to the potential difference across the ohmic resistance at frequency ..... . *(Exp. June 21)*



(a) c only (b) b and d

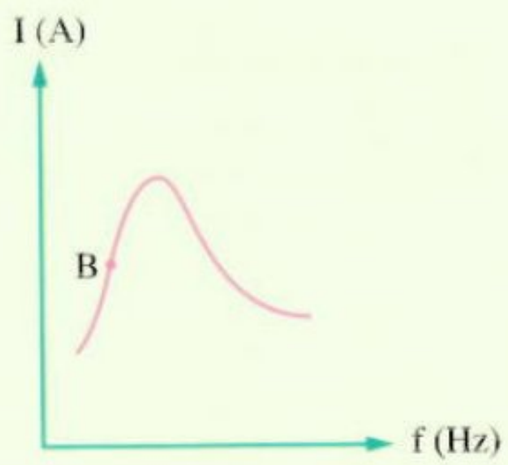
(c) a only (d) a and c

3 What change should be done to the self-inductance of the coil in an oscillating circuit to double the frequency of the current? *(1st session 21)*

(a) Increase it to its double. (b) Increase it to its quadruple.

(c) Decrease it to its quarter. (d) Decrease it to its half.

4 An alternating current circuit contains an inductive coil of negligible ohmic resistance, a capacitor of variable capacitance and an ohmic resistance which are connected in series. From the opposite graph, the ratio between the source voltage and the potential difference across the ohmic resistance at point (B) ..... . *(1st session 21)*

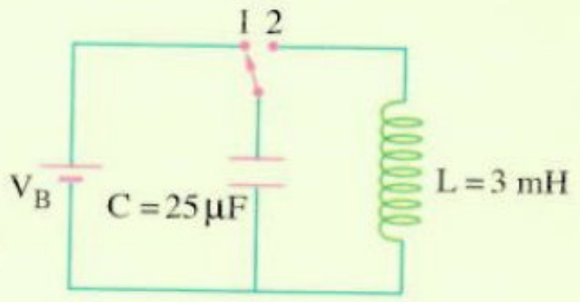


(a) is less than one (b) equals zero

(c) equals one (d) is greater than one

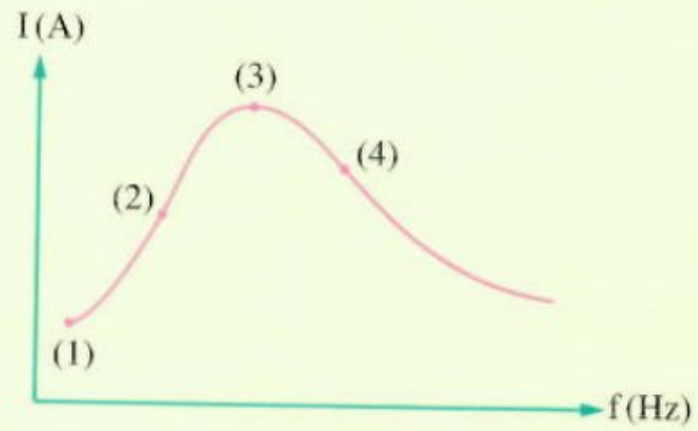
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5 The figure shows an oscillating circuit containing capacitor of capacitance C and inductive coil of inductance L, then the value of the frequency for the alternating current in the oscillating circuit when the key is switched from position (1) to position (2) equals ..... . ($\pi = 3.14$) *(2nd session 21)*



(a) 0.58 Hz (b) 0.0183 Hz (c) 58.14 Hz (d) 581.4 Hz

6 An alternating current circuit contains an inductive coil of negligible ohmic resistance, a capacitor of variable capacitance and an ohmic resistance connected together in series. From the shown graph, the resultant of inductive reactance and capacitive reactance vanishes at point ..... . *(2nd session 21)*



(a) 1 (b) 2 (c) 3 (d) 4

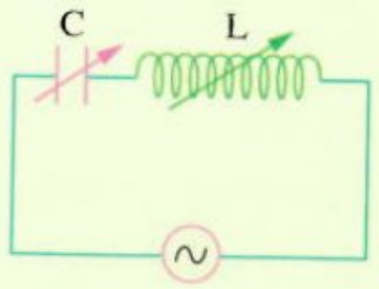
7 A tuning circuit (X) consists of an inductor with inductance 0.2 H and a capacitor of capacitance 0.2 μF. Another tuning circuit (Y) consists of an inductor of self inductance 0.4 H and a capacitor of capacitance 0.1 μF, then the ratio of :

$$\frac{\text{the resonance frequency in circuit (X)}}{\text{the resonance frequency in circuit (Y)}} = ...$$

(1st session 22)

(a) $\frac{2}{1}$ (b) $\frac{1}{4}$ (c) $\frac{1}{1}$ (d) $\frac{4}{1}$

8 The figure shows a series tuning circuit that consists of variable capacitor and inductor with ohmic resistance. When the capacitance of the capacitor is increased to the double with keeping the same resonance frequency, then the ratio between the inductive reactance in first case to that in second case $\frac{X_{L1}}{X_{L2}} = ...$ *(2nd session 22)*



(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{4}{1}$ (d) $\frac{2}{1}$

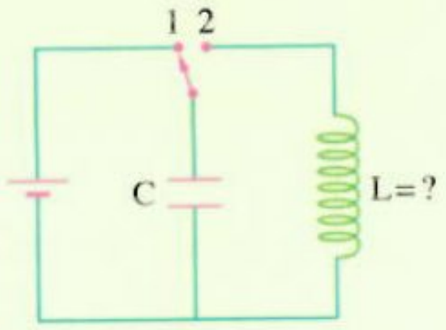
9 Resonant circuit of frequency 2×10^{14} Hz has a capacitor of capacitance (C) farad and a coil of self-induction (L) Henry, if the capacitance is increased to (9 C) farad and self-inductance of the coil decreases to $\left(\frac{L}{9}\right)$ Henry, then the resonate frequency will ..... .

(a) increase three times (b) remain constant
(c) increase 9 times (d) decrease to third *(Exp. 23)*

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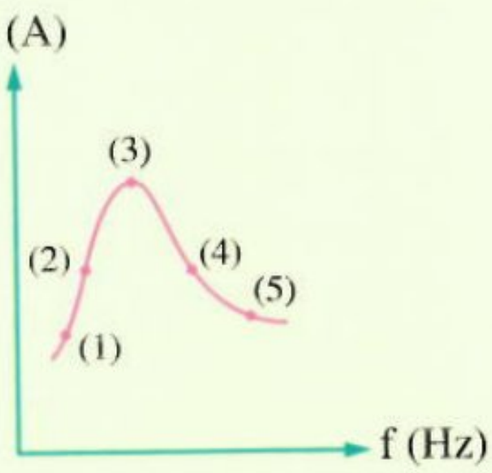
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10 The figure shows an oscillatory circuit containing a capacitor of capacitance 200 μF. What is the value of the inductance of the coil (L) required to obtain a current of frequency that equals 100 Hz when switching the key from position (1) to (2)? ($\pi = 3.14$) *(1st session 23)*



(a) 12.68 henry (b) 0.0127 henry
(c) 78.75 henry (d) 1.267×10^{-8} henry

11 The graph of effective current (I) versus frequency (f) for an alternating current circuit in which a non-inductive ohmic resistor, an induction coil of negligible ohmic resistance and a capacitor of variable capacitance are connected in series is shown in the opposite figure, so in the figure, the points where the potential difference between the two plates of the capacitor is greater than the potential difference between the two ends of the coil are ....., ... *(1st session 23)*

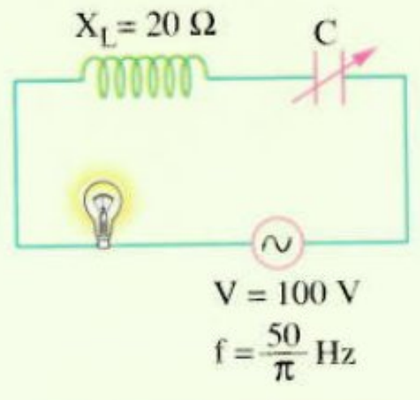


(a) points (2), (3) (b) points (4), (5)
(c) points (1), (2) (d) points(2), (4)

12 An oscillating circuit consists of a coil of inductance 1H and a capacitor of 3.5 μF, so the frequency of the oscillating circuit current is ..... . (Take: $\pi = 3.14$) *(2nd session 23)*

(a) 45.495 kHz (b) 85.1 Hz (c) 0.085 Hz (d) 13.55 Hz

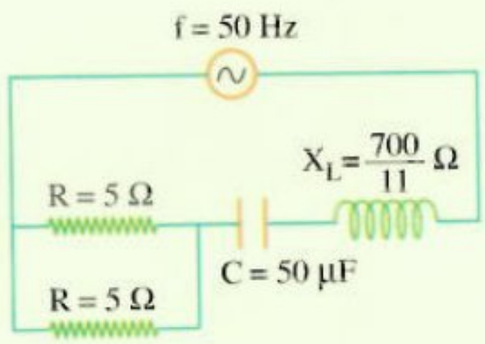
13 According to the given figure, the capacitance of the capacitor (C) which causes maximum lightning of the electric lamp equals ..... farad. *(2nd session 23)*



(a) 2.5×10^{-4} (b) 10×10^{-4}
(c) 5×10^{-4} (d) 15×10^{-4}

14 In the AC circuit in the figure, the total potential difference for the circuit ..... .

(Where: The source frequency = 50 Hz, $\pi = \frac{22}{7}$) *(2nd session 23)*



(a) lags the current by 90°
(b) leads the current by 45°
(c) lags the current by 45°
(d) and the current has the same phase

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15 What change would take place to the current frequency in an oscillating circuit if each of the self-inductance of its coil and the capacitance of the capacitor are doubled? *(1st session 24)*

(a) Increases to four times.
(b) Decreases to its quarter.
(c) Decreases to its half.
(d) Increases to double.

16 An oscillating circuit contains a capacitor and an inductor of inductance $\frac{1}{5}$ H. To increase the frequency of the circuit current to the double, we should add another inductor in parallel of inductance that is equal to ..... . *(2nd session 24)*

(a) $\frac{1}{25}$ H (b) $\frac{1}{15}$ H (c) $\frac{3}{20}$ H (d) $\frac{1}{5}$ H

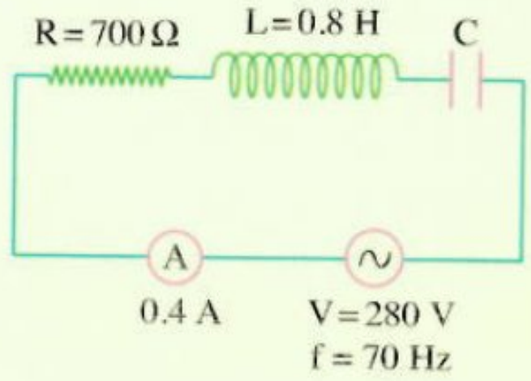
17 In an RLC series circuit containing an ohmic resistance R, an inductive reactance $(X_L)_1$ of negligible ohmic resistance, and a capacitive reactance $(X_C)_1$, the circuit is at resonance. When the capacitor is replaced with another capacitor whose capacitive reactance is $(X_C)_2$, it is necessary to double the inductance of the coil in order to maintain resonance state in both cases. The ratio of $\left(\frac{(X_C)_1}{(X_C)_2}\right)$ is equal to ..... . *(2nd session 24)*

(a) $\frac{2}{1}$ (b) $\frac{4}{1}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

18 In an RLC circuit, which of the following plays a role in determining the resonance frequency of the circuit? *(Practice 25)*

(a) The total ohmic resistance of the circuit.
(b) The maximum value of the applied AC voltage to the circuit.
(c) The total capacitance and the total inductance of the circuit.
(d) The total reactance of the circuit.

19 In the electric circuit shown in the figure, neglecting the ohmic resistance of the hot-wire ammeter, the capacitance of the capacitor is ..... . *(Practice 25)*



(a) 4.24 μF (b) 5.68 μF
(c) 6.46 μF (d) 8.72 μF

20 In an RLC, series circuit, if $V_R = V_L = V_C = V_{source}$, so when removing the inductor from the circuit, the voltage across resistor R becomes ..... . *(Practice 25)*

(a) $\sqrt{2}\, V_{source}$ (b) $\frac{V_{source}}{\sqrt{2}}$ (c) V_{source} (d) $\frac{V_{source}}{2}$

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21 A radio receiver contains a tuning circuit having a changeable capacitor and a constant inductor for receiving different radio stations. If the self-inductance of the inductor used in the tuning circuit is 1 mH, and the required frequency range is from 540 kHz to 1600 kHz, then the approximate range of the required capacitance should be ..... . *(Practice 25)*

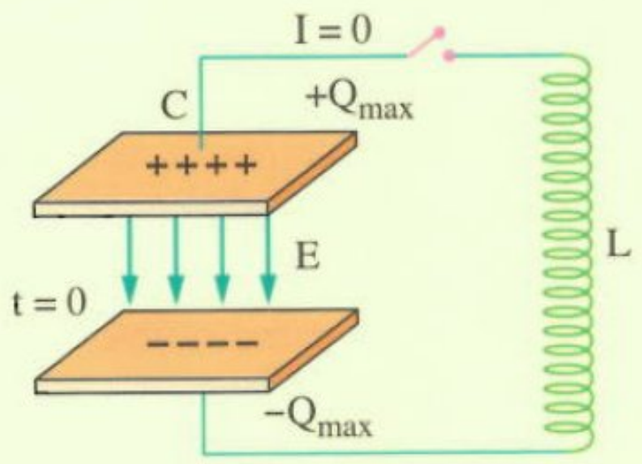
(a) from 10 pF to 87 pF
(b) from 100 pF to 870 pF
(c) from 1 nF to 8.7 nF
(d) from 10 nF to 87 nF

22 Electromagnetic waves resulting from an oscillating circuit are damped due to ..... . *(Practice 25)*

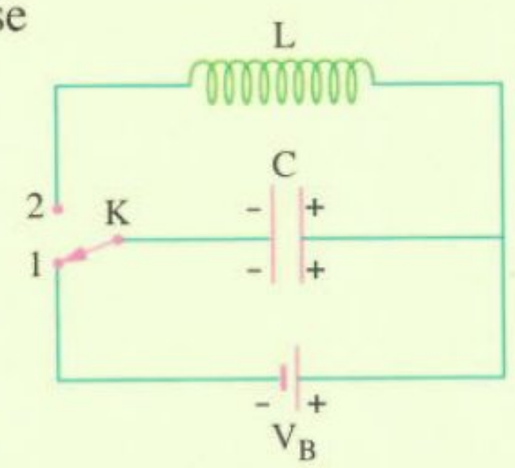
(a) the ohmic resistance of the circuit
(b) the inductive reactance of the coil
(c) the capacitive reactance of the circuit
(d) the total reactance of the circuit

23 In the figure in front of you, after closing the switch and during a quarter cycle of oscillation, the induced electromotive force in the coil will be ..... . *(Practice 25)*

(a) back induced
(b) constant in magnitude
(c) increases with time
(d) forward induced

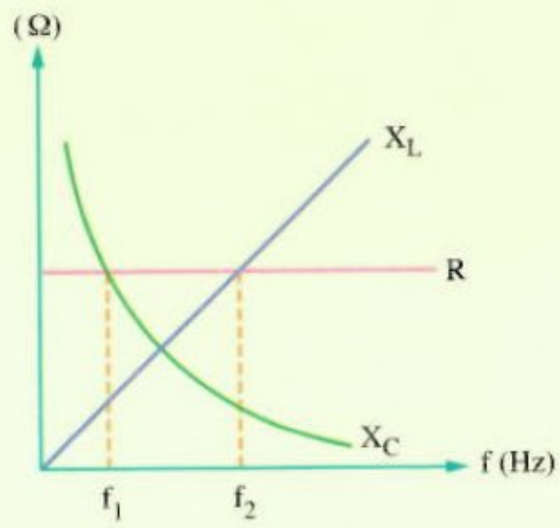


24 ✱ The figure shows an oscillating circuit containing a coil whose self-inductance is (L) and a capacitor of capacitance (C), each of its plates is charged with a quantity of charge (q). Starting from the instant the switch (K) is moved to position (2), what is the shortest interval of time required until each capacitor plate is charged with an opposite charge of magnitude (q)? *(1st session 25)*
(Knowing that: the periodic time is the reciprocal of frequency)



(a) $\pi\sqrt{LC}$
(b) $2\pi\sqrt{LC}$
(c) $\frac{1}{2\pi\sqrt{LC}}$
(d) πLC

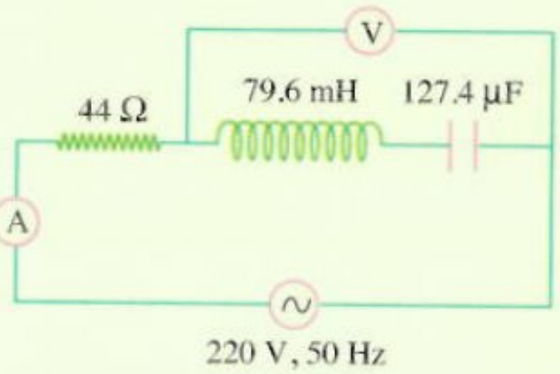
25 The graph shows the effect of changing the frequency of an AC source in (LCR) circuit on each of the inductive reactance, the capacitive reactance, and the ohmic resistance.
What is the phase difference between the total voltage and current in the circuit at each of the two frequencies f_1 and f_2? *(1st session 25)*



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	At frequency f_1	At frequency f_2
a	The total voltage and current are in the same phase	The total voltage and current are in the same phase
b	The total voltage leads the current by an angle θ	The total voltage lags the current by an angle θ
c	The total voltage lags the current by an angle θ	The total voltage leads the current by an angle θ
d	The total voltage leads the current by an angle θ	The total voltage leads the current by an angle θ

26 The figure shows LCR circuit containing an AC source (220 V and 50 Hz), a capacitor of capacitance 127.4 μF, a coil of inductance 79.6 mH and negligible ohmic resistance, all are connected in series.

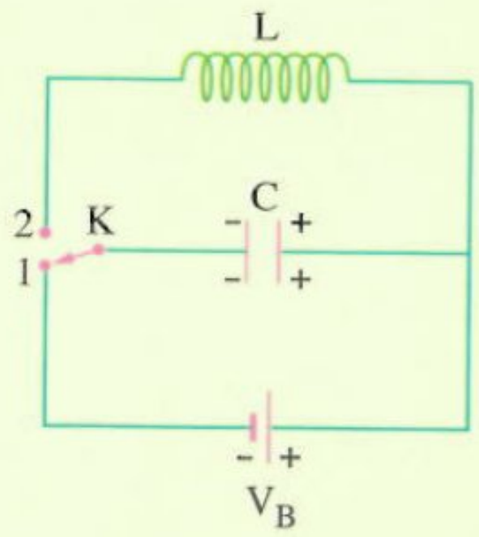


If the ohmic resistance in the circuit is 44 Ω, and ($\pi = 3.14$), the voltmeter and ammeter readings are ..... .

(1st session 25)

	Voltmeter reading	Ammeter reading
a	150 V	2 A
b	150 V	6 A
c	0 V	5 A
d	0 V	4 A

27 The figure shows an oscillating circuit containing an inductive coil (L) and a capacitor (C), each of its two plates is charged with a charge (q). Starting from the moment the switch (K) comes in contact with position (2), what is the shortest time interval that passes until all the electrical energy in the capacitor is completely converted into magnetic energy in the coil?



(Note that: Periodic time is the reciprocal of frequency)

(a) πLC (b) $\pi\sqrt{LC}$

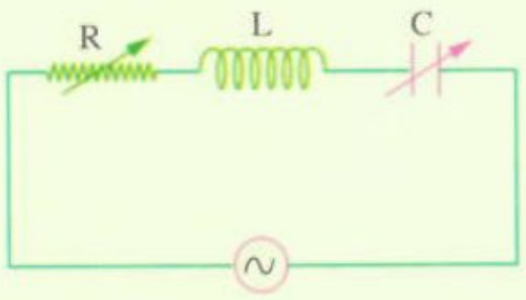
(c) $2\pi\sqrt{LC}$ (d) $\frac{\pi}{2}\sqrt{LC}$

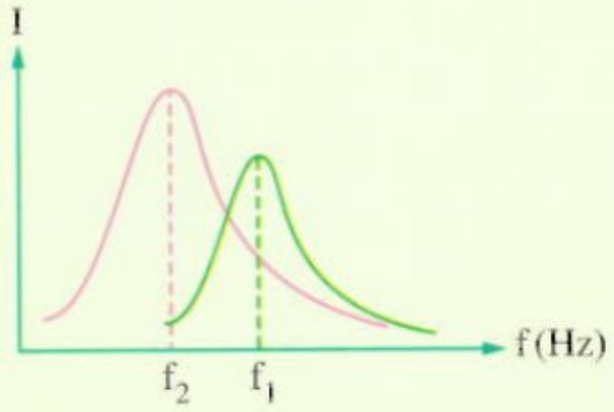
(2nd session 25)

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28 An alternating current circuit (LCR) contains AC supply, an inductor (L), a capacitor (C) of variable capacitance, and a variable resistor (R). The graph shows the relationship between the effective value of current (I) and the current frequency (f). The circuit was in resonance at frequency (f_1),



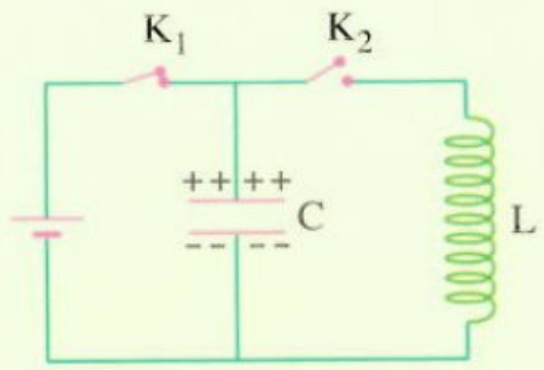


what is the required change in each of the capacitor capacitance and the ohmic resistance to bring the circuit into resonance again at frequency (f_2) as represented by the curve in the given graph?

	Capacitor capacitance	Ohmic resistance
(a)	Increases	Increases
(b)	Increases	Decreases
(c)	Decreases	Increases
(d)	Decreases	Decreases

(2nd session 25)

29 The figure shows an oscillating circuit containing a capacitor (C) of capacitance $\frac{25}{\pi}\ \mu F$ and an inductor of self-inductance $\frac{1}{\pi}$ H. If switch K_1 is opened and switch K_2 is closed, then after a time interval of 5 ms from the moment switch K_2 is closed, the capacitor (C) becomes ..... . (Assume there is no energy loss)



(Practice 26)

(a) uncharged
(b) partially charged
(c) fully charged with an opposite charge
(d) fully charged with the same charge

30 Two oscillating circuits (X, Y) contain two identical inductors and capacitors, the resistance of the wires of circuit (X) was R_X and the resistance of the wires of circuit (Y) was R_Y. If the damping of the charge on the plates of the capacitor in circuit (X) is faster than that on plates of the capacitor in circuit (Y), the choice that correctly expresses the relation between resistors R_X and R_Y is ..... . *(Practice 26)*

(a) $R_X > R_Y$
(b) $R_X = R_Y$
(c) $R_X < R_Y$
(d) the answer cannot be determined

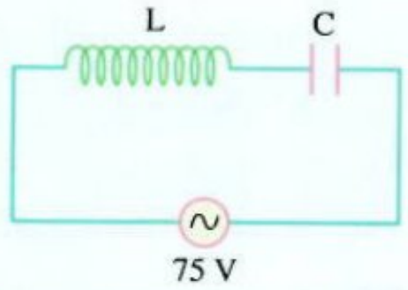
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Test on Chapter 4

Answered

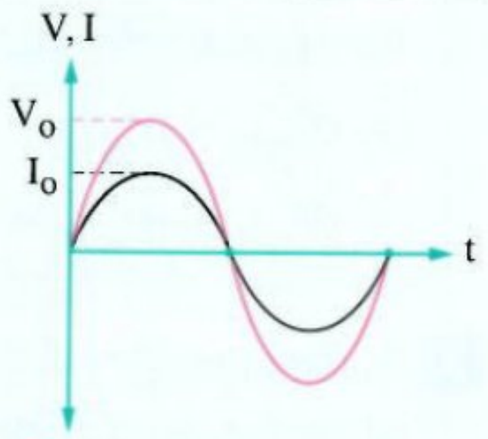
First Choose the correct answer

1 In the opposite electric circuit, if the ohmic resistance in the circuit is negligible, the voltages across the inductor and across the plates of the capacitor could be ..... .



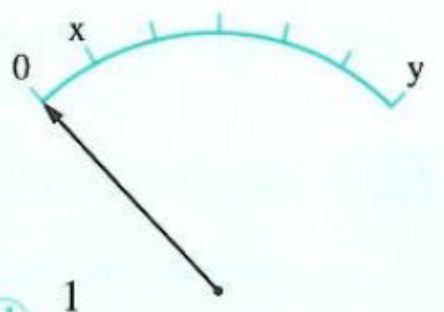
(a) 200 V, 125 V (b) 50 V, 25 V (c) 100 V, 150 V (d) 75 V, 75 V

2 An AC source is connected to a pure component. The opposite graph represents the relation between each of the current intensity (I) through the circuit and the voltage (V) across the source versus time (t), so the electrical power consumption in the circuit equals ..... .



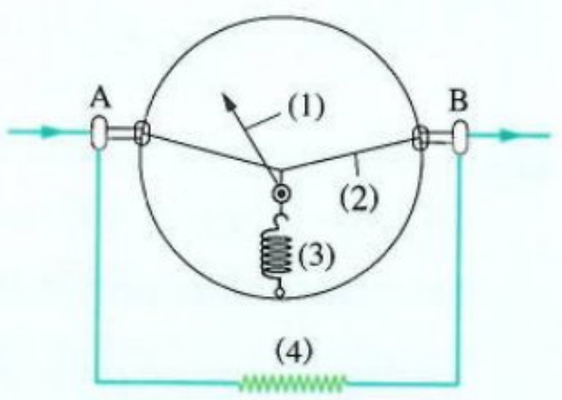
(a) $I_o V_o$ (b) $\frac{I_o V_o}{\sqrt{2}}$ (c) $\frac{I_o V_o}{2}$ (d) zero

3 The opposite figure illustrates equal divisions on the scale of a hot-wire ammeter, so the ratio between the readings of the hot-wire ammeter at positions x , y $\left(\frac{I_x}{I_y}\right)$ equals ..... .



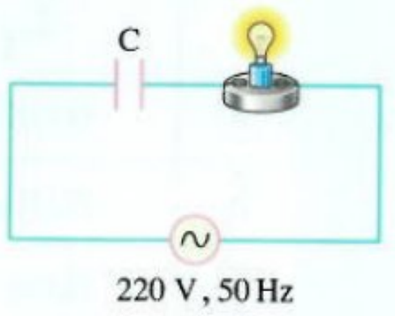
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{6}}$

4 The opposite figure illustrates the structure of one of the electrical measuring devices, so the component whose length decreases when an electric current passes through the device is ..... .



(a) (1) (b) (2)
(c) (3) (d) (4)

5 The opposite electric circuit illustrates an AC source (220 V, 50 Hz) connected to an electric lamp (110 V, 55 W) and a capacitor C, so the required capacitance of the capacitor for the lamp to operate at its full power is ..... .



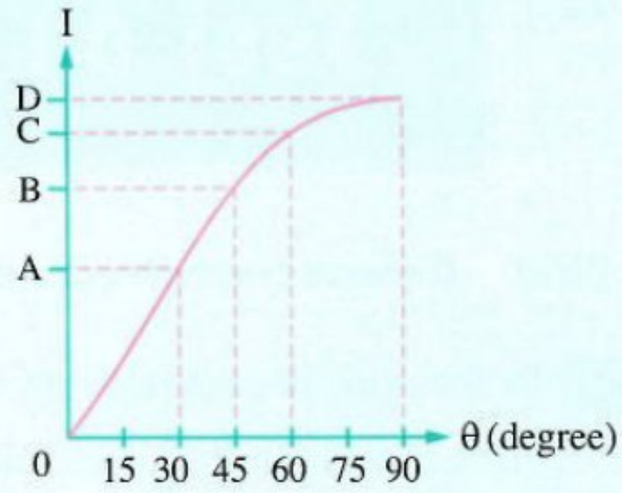
(a) 1.21 μF (b) 3.52 μF (c) 6.24 μF (d) 8.35 μF

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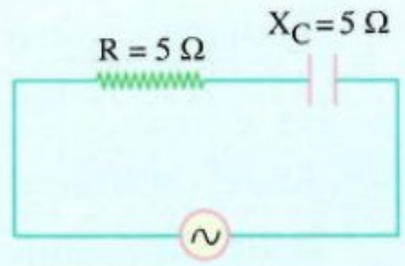
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6 The opposite graph represents one quarter of a cycle of an alternating current. Which of the values of current shown on the graph will be the reading of a hot-wire ammeter when this current passes through it?



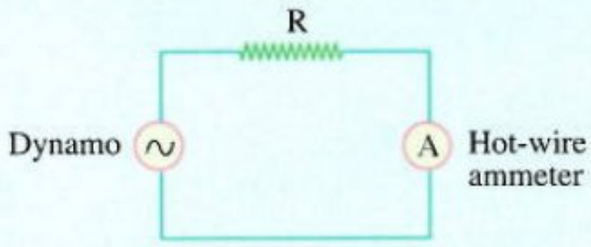
(a) Value A (b) Value B
(c) Value C (d) Value D

7 Which of the following options is correct regarding the power consumed in the two components of the opposite electric circuit?



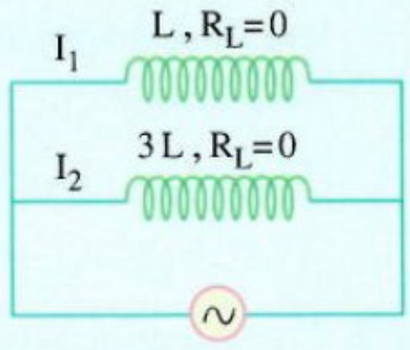
(a) $(P_w)_R < (P_w)_C$ (b) $(P_w)_R = (P_w)_C$
(c) $(P_w)_R > (P_w)_C \neq 0$ (d) $(P_w)_R > (P_w)_C = 0$

8 In the opposite AC circuit, what happens to the reading of the hot-wire ammeter after some time of doubling the angular speed of the dynamo coil?



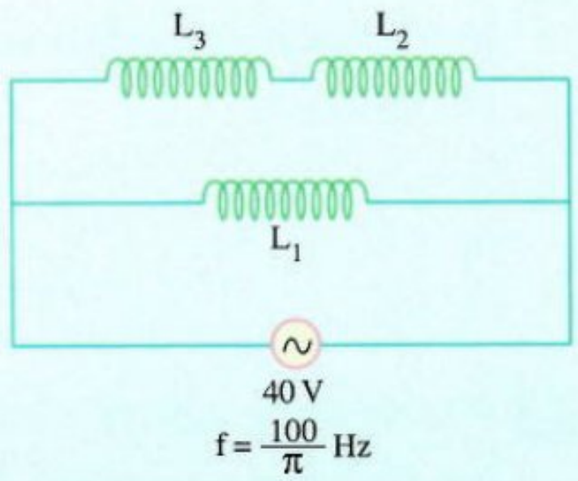
(a) Doubles. (b) Halves.
(c) Decreases to its quarter. (d) Remains unchanged.

9 In the opposite electric circuit, the ratio between the intensities of currents through the two coils $\left(\frac{I_1}{I_2}\right)$ equals ..... .



(a) $\frac{\sqrt{3}}{1}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\frac{3}{1}$ (d) $\frac{1}{3}$

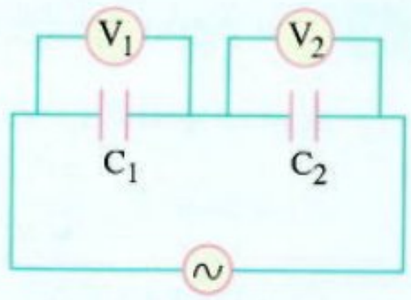
10 In the opposite electric circuit, an alternating current of effective value 5 A flows through the source. If the inductors have negligible ohmic resistance and the self-inductance of inductor L_3 is 0.02 H, then the possible values for the self-inductances of inductors L_1 and L_2 are ..... , ...

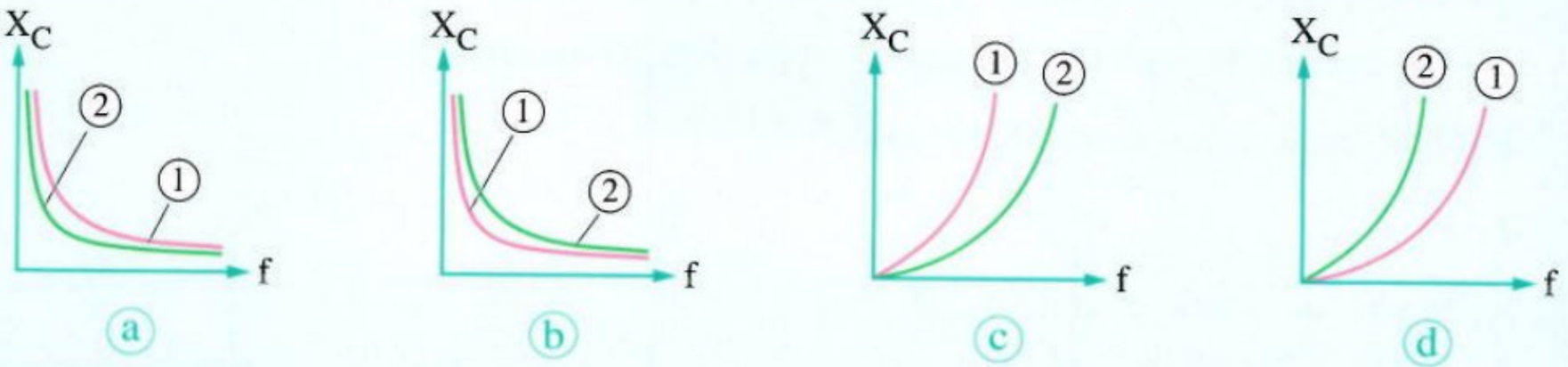


	L_1	L_2
(a)	0.08 H	0.06 H
(b)	0.06 H	0.08 H
(c)	0.04 H	0.06 H
(d)	0.08 H	0.04 H

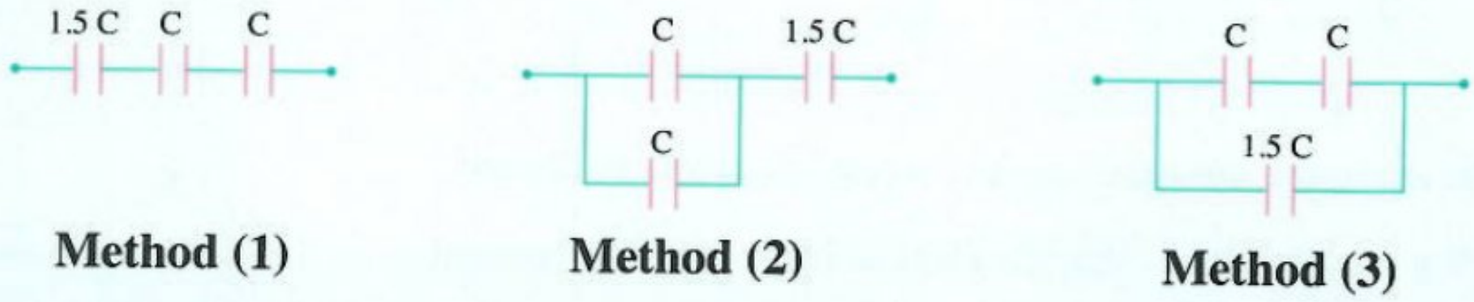
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11 In the opposite electric circuit, if the reading of voltmeter V_1 is greater than the reading of voltmeter V_2, which of the following graphs represents the relation between the capacitive reactance (X_C) for each capacitor and the frequency of the electric source (f)?





12 Three capacitors are connected together in three different methods (1), (2), and (3) as shown in the following figures:



Method (1) **Method (2)** **Method (3)**

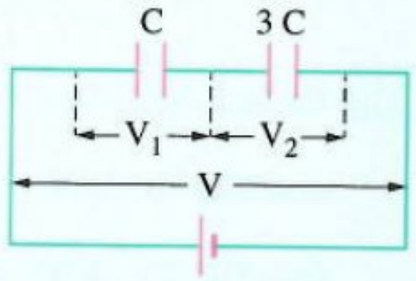
So, the correct order of these methods according to the equivalent capacitance of the three capacitors is ..... .

(a) 1 > 2 > 3 (b) 3 > 2 > 1 (c) 3 > 1 > 2 (d) 2 > 3 > 1

13 In the opposite electric circuit, the ratio of voltages ($V : V_1 : V_2$) respectively equals ..... .

(a) 4 : 1 : 3 (b) 4 : 3 : 1

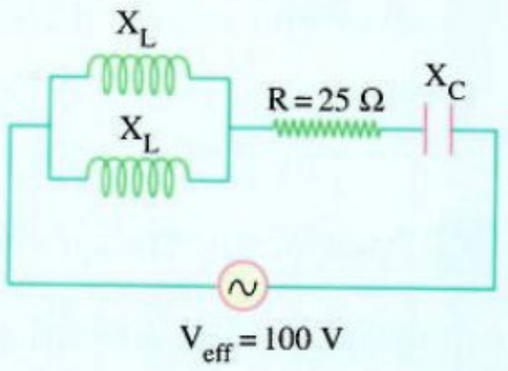
(c) 4 : 2 : 2 (d) 3 : 2 : 5



14 In the opposite electric circuit, two inductors have negligible ohmic resistance, if the intensity of current through each of them is 2 A, and the potential difference across the terminals of the capacitor is 100 V, then the value of X_L equals ..... .

(a) 25 Ω (b) 50 Ω

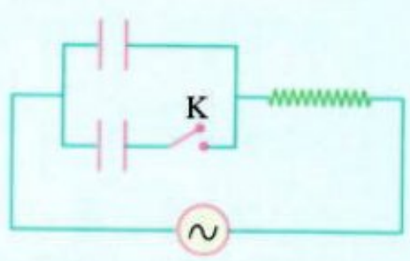
(c) 100 Ω (d) 200 Ω



15 What happens to the phase angle between the total voltage and the current through the opposite electric circuit when closing switch K?

(a) Increases. (b) Decreases.

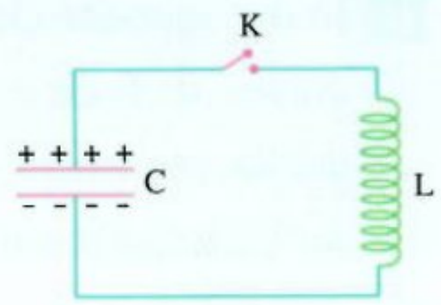
(c) Doesn't change. (d) Becomes zero.

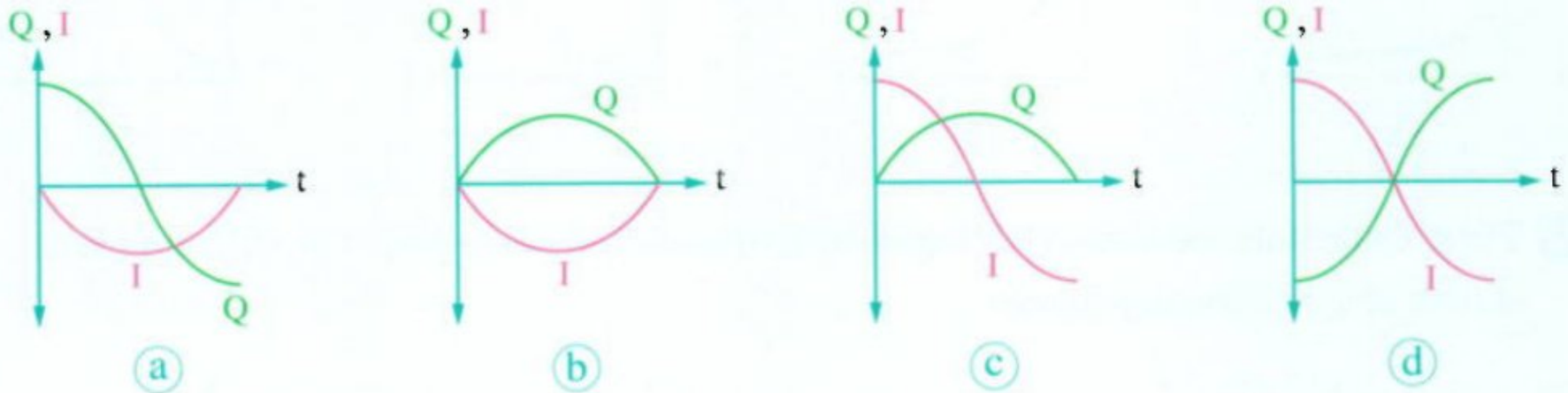


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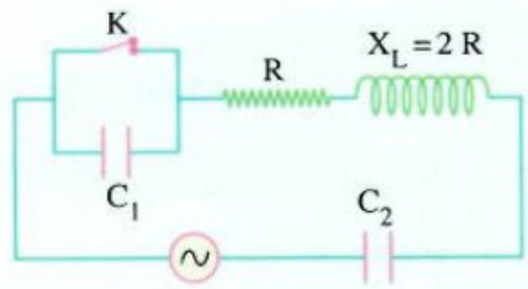
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16 The opposite figure illustrates an electric circuit in which a charged capacitor C is connected to an inductor L through a switch K, so which of the following graphs represents the changes in both of the electric charge on the capacitor plates (Q) and the current through the inductor (I) with time (t) during the first half-cycle when switch K is closed?



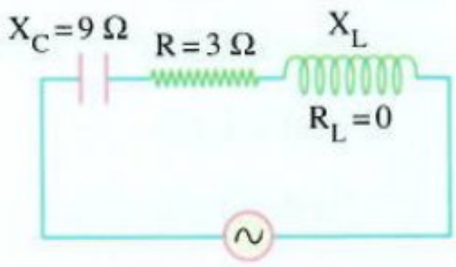


17 In the opposite electric circuit, when switch K is closed, the phase angle between the total voltage and the current equals 45° and when switch K is opened, the phase angle becomes – 45°, so the ratio $\left(\frac{C_2}{C_1}\right)$ equals ..... .



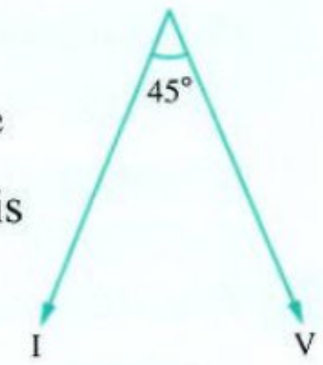
(a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{1}{2}$ (d) $\frac{2}{1}$

18 In the opposite electric circuit, if the total voltage across the source leads the current by an angle of 45°, the inductive reactance of the inductor (X_L) equals ..... .



(a) 3 Ω (b) 6 Ω
(c) 9 Ω (d) 12 Ω

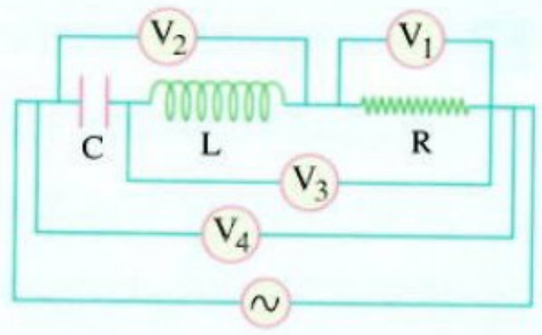
19 An electric circuit consists of an AC source, an inductor (L) of an ohmic resistor (R), and the opposite figure represents the phasors of total voltage (V) and current (I) in the circuit, so the total impedance (Z) of the circuit is given by the relationship .....



(a) $Z = 2R$ (b) $Z = 2X_L$
(c) $Z = \sqrt{2}R$ (d) $Z = R^2 + X_L^2$

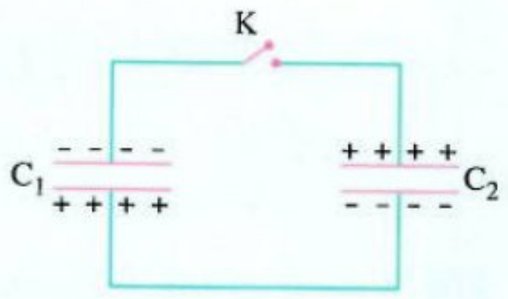
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20 In the opposite electric circuit, the components R, L, and C are pure and in a resonant state, so which two voltmeters among the shown voltmeters show the same reading?



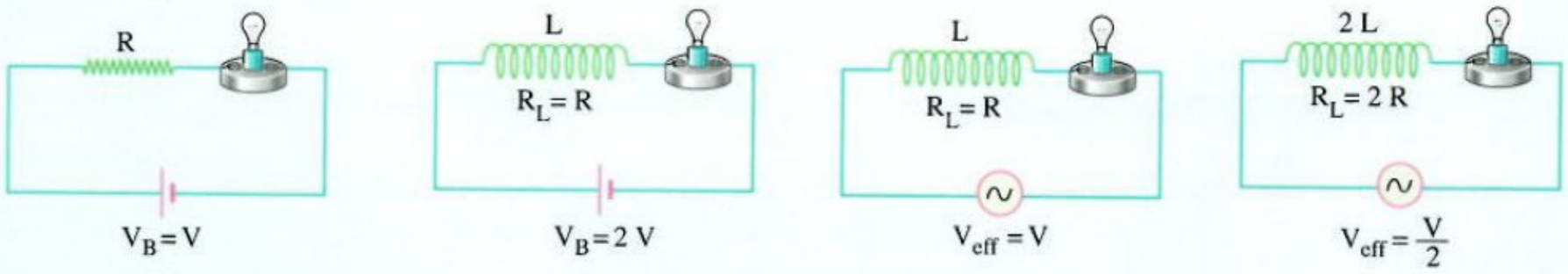
(a) V_1 and V_2 (b) V_2 and V_3
(c) V_3 and V_4 (d) V_1 and V_4

21 The opposite figure shows an electric circuit consisting of two capacitors, C_1 and C_2, of capacitances of 1 μF and 4 μF respectively, and the potential differences across them are 5 V and 10 V respectively. If switch K is closed, the potential difference across capacitor C_1 and the potential difference across capacitor C_2 respectively, become ..... .



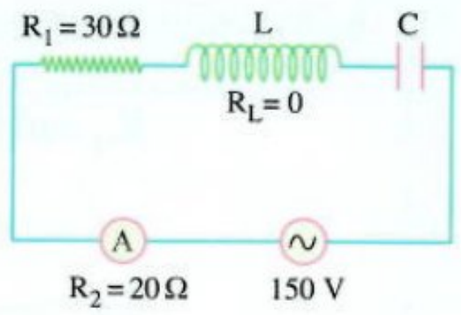
(a) 7 V , 8 V (b) 8 V , 7 V (c) 7 V , 7 V (d) 7.5 V , 7.5 V

22 A light bulb of resistance R is connected in different electric circuits with different electric sources of negligible internal resistance as follows, so in which of the following circuits is the brightness of the light bulb the lowest?



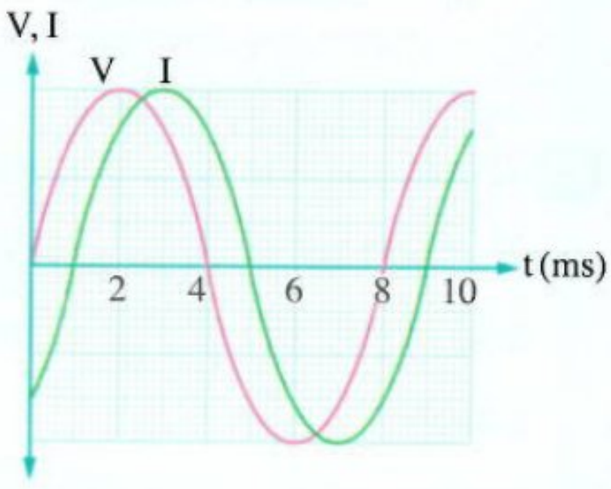
(a) (b) (c) (d)

23 When only the inductor is removed from the opposite circuit, the total voltage lags the current by an angle of 52° and when only the capacitor is removed from the opposite circuit, the total voltage leads the current by an angle of 52°, so the reading of the hot-wire ammeter in the opposite circuit equals ..... .



(a) 3 A (b) 2.75 A (c) 2.25 A (d) 1.75 A

24 An AC circuit consists of an alternating current source, an ohmic resistor (R), and an inductor (X_L) of negligible ohmic resistance. The opposite graph represents the variation of the total voltage (V) and the current (I) versus time (t), so the ratio $\left(\frac{X_L}{R}\right)$ equals ..... .

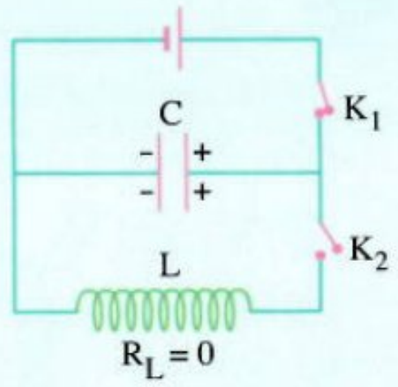


(a) $\frac{1}{9}$ (b) $\frac{1}{8}$
(c) $\frac{1}{4}$ (d) $\frac{1}{1}$

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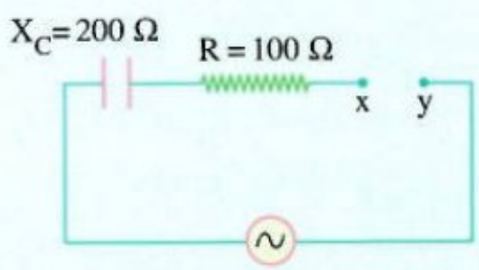
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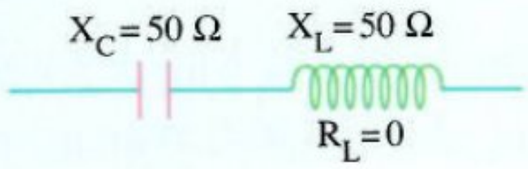
25 In the opposite electric circuit, a capacitor of capacitance C carrying an electric charge on one of its plates Q is connected to an inductor of self-inductance L. If switch K_1 is opened and then switch K_2 is closed, the time taken for the plate of the capacitor to get fully charged with an opposite charge to its initial charge for the first-time equals ..... .

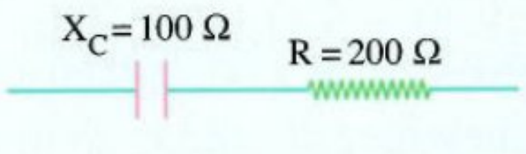


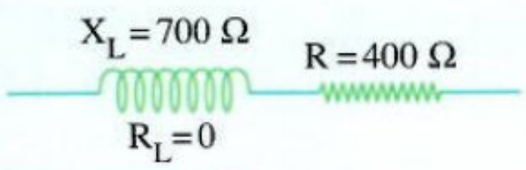
(a) $\pi\sqrt{LC}$ (b) $2\pi\sqrt{LC}$ (c) $\frac{\pi}{2}\sqrt{LC}$ (d) $\frac{\pi}{4}\sqrt{LC}$

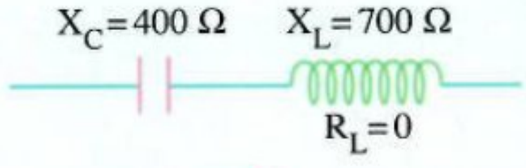
26 Which of the following electrical components can be connected between points x and y in the opposite electric circuit so that the total impedance of the circuit does not change?



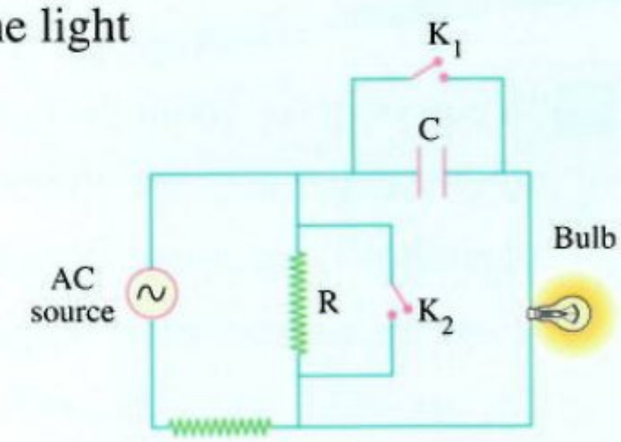
(a) 

(b) 

(c) 

(d) 

27 In the opposite circuit, what happens to the brightness of the light bulb when closing switch:



	K_1 only?	K_2 only?
(a)	Increases	Doesn't change
(b)	Increases	Vanishes
(c)	Doesn't change	Decreases
(d)	Doesn't change	Doesn't change

28 The instantaneous electromotive force induced in an AC dynamo is given by the equation ; $V = 200 \sin 18000t$, so if the dynamo is connected to an inductor of inductance 0.1 H and negligible ohmic resistance, the effective value of the current through the coil equals ..... .

(a) 0.08 A (b) 0.11 A (c) 4.5 A (d) 6.4 A

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